

Object Oriented Design And Programming

Sem 2

Watermarked study resource

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OODP Important Topic

UNIT-1

1. Basic concept of OOPs

2. Features of OOPs

3. C++ Access Specifiers

4. Write a C++ program for

☐ Reverse a numbers

☐ Fibonacci series

☐ Armstrong number or not

☐ prime number or not

☐ sum of even numbers

☐ sum of odd numbers

5. Draw a Use case diagram for any one application (Some time they will ask this question as a scenario based Question)

☐ Banking Management System

☐ ATM Management System

☐ Online Ticket Reservation Management System

☐ Online Credit Card Processing System

☐ Online Course Reservation System

☐ Online purchasing System

☐ Stock management System

☐ Restaurant Management System

☐ Library Management System

☐ Hospital Management System

6. Draw a Class diagram for any one application (Some time they will ask this question as a scenario based Question)

☐ Banking Management System

☐ ATM Management System

☐ Online Ticket Reservation Management System

☐ Online Credit Card Processing System

☐ Online Course Reservation System

☐ Online purchasing System

- ☒ Stock management System
- ☒ Restaurant Management System
- ☒ Library Management System
- ☒ Hospital Management System

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UNIT-2

1. Types of constructors with an example
2. Destructor with an example
3. Method Overloading with an example
4. Operator Overloading with an example
5. Draw a UML Interaction Diagrams (Sequence Diagram / Collaboration Diagram) any one application (Some time they will ask this question as a scenario based Question)

- ☒ Banking Management System
- ☒ ATM Management System
- ☒ Online Ticket Reservation Management System
- ☒ Online Credit Card Processing System
- ☒ Online Course Reservation System
- ☒ Online purchasing System
- ☒ Stock management System
- ☒ Restaurant Management System
- ☒ Library Management System
- ☒ Hospital Management System

UNIT-3

1. Multiple Inheritance with an example
2. Multilevel inheritance with an example
3. Hierarchical with an example
4. Hybrid Inheritance with an example
5. Friend function with an example
6. Virtual function with an example
7. Pure Virtual function with an example
8. Abstract class with an example
9. Interface or Pure abstract class with an example
10. Draw UML State Chart Diagram for any one application (Some time they will ask this question as a scenario based Question)

- ☒ Banking Management System
- ☒ ATM Management System
- ☒ Online Ticket Reservation Management System

- ☐ Online Credit Card Processing System
- ☐ Online Course Reservation System
- ☐ Online purchasing System
- ☐ Stock management System
- ☐ Restaurant Management System
- ☐ Library Management System
- ☐ Hospital Management System

11. Draw UML Activity Diagram for any one application (Some time they will ask this question as a scenario based Question)

- ☐ Banking Management System

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- ☐ ATM Management System
- ☐ Online Ticket Reservation Management System
- ☐ Online Credit Card Processing System
- ☐ Online Course Reservation System
- ☐ Online purchasing System
- ☐ Stock management System
- ☐ Restaurant Management System
- ☐ Library Management System
- ☐ Hospital Management System

Unit-4

1. Function template
 2. Class template
 3. Exception handling (try, throw, catch and multiple catch)
 4. Types of exceptions
 5. Package diagram , Component diagram and Deployment diagram for following applications (Some time they will ask this question as a scenario based Question)
- ☐ Banking Management System
 - ☐ ATM Management System
 - ☐ Online Ticket Reservation Management System
 - ☐ Online Credit Card Processing System
 - ☐ Online Course Reservation System
 - ☐ Online purchasing System
 - ☐ Stock management System
 - ☐ Restaurant Management System
 - ☐ Library Management System
 - ☐ Hospital Management System

UNIT-5

1. Sequence Container: Vector, List, Deque, Array, Stack with an example.
2. Associative Containers: Map, Multimap with an example.

3. Algorithms: find(), count(), sort(), search(), merge(), for_each(), transform() with an example.

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

RAMAPURAM CAMPUS, CHENNAI-89

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

18CSC202J-OBJECT ORIENTED DESIGN AND PROGRAMMING

QUESTION BANK

UNIT-1

1 Marks

1. Which of the following explains Polymorphism? (L1) (Page no 19)

A) intfunc(int, int);

Float func1(float,float);

B) intfunc(int);

Intfunc(int);

C) intfunc(float);

Intnew_func();

D) intfunc();

Intnew_func();

Ans: C

2. Find how many bytes are occupied by the following data types in a 32-bit system. (L1)

(Page no 70)

Reference-5

A) Type int

B) Type long double

C) Type float

D)Type long

Ans: C

3. Which of the following is a important role of a function? (L1) (Page no 210)

A) give a name to a block of code

B) reduce program size

C) accept arguments and provide a return value

D) help organize a program into conceptual units

Ans:D

4. The Unified Modeling Language is(L1) (Page no 27)- Reference-5

- A) a program that builds physical models.
- B) a way to look at the organization of a program
- C) the combination of C++ and FORTRAN
- D) helpful in developing software systems.

Ans:D

5. In C++, a function contained within a class is called(L1) (Page no 25)

- A) a member function

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- B) an operator
- C) a class function
- D) a method

Ans:A

6. What happens if the base and derived class contains definition of a function with same prototype?

(L1)(Page no 21)

- A) Compiler reports an error on compilation
- B) Only base class function will get called irrespective of object
- C) Only derived class function will get called irrespective of object
- D) Base class object will call base class function and derived class object will call derived class object will call derived class function

Ans:D

7. Which one of the following option is correct about the statement given below? The compiler checks the type of reference in the object and not the type of object(L1) (Page no 29)

- A) inheritance
- B) Polymorphism
- C) Abstraction
- D) Encapsulation

Ans:B

8. Which of the following functions are performed by a constructor? (L1)(Page no 32)

- A) Construct a new class
- B) Construct a new object
- C) Construct a new function
- D) Initialize objects

Ans: D

9. Which of the following is the correct class of the object cout(L1) (Page no 38)

- A) iostream

- B) istream
- C) ostream
- D) ifstream

Ans: C

10. In UML, diagrams which captures system static structure and provide foundation for other models is

called (L1) (Page no 35)

- A) Deployment diagram
- B) Class diagram
- C) Component diagram
- D) Object diagram

Ans: B

11. Find the error produced by compiler when private members are accessed? (L1) (Page no 31)

- A) Can't access private message
- B) Code unreachable
- C) Core dumped
- D) Bad code

Ans: A

12. Choose the default access specifier for the class member(L1) (Page no 40)

- A) public

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- B) private
- C) protected
- D) None of the above

Ans: B

13. Which of the following is CPP style type-casting? (L1) (Page no 42)

- A) per=total/ (float)m
- B) per=total/float(m)
- C) per = (float)total/m
- D) None of these

Ans: B

14.What is the output of the following program? (L1) (Page no 37)

```
#include<iostream>
using namespace std;
void main()
{
char s() = "SRM";
*s = 'R';
cout<<s<<endl;
```

}

A) RRM B) SRM C) SRR D) None of these

Ans: A(Page no 37)

15.What does the following statement mean? (L1) (Page no 42)

int (*fp)(char*)

A) pointer to a pointer

B) pointer to an array of chars

C) pointer to function taking a char* argument and returns an int

D) function taking a char* argument and returning a pointer to int

Ans : C

16.Which of the following concepts of OOPS means exposing only necessary information to client? (L1) (Page

no 42)

A) Encapsulation

B) Abstraction

C) Polymorphism

D) Data binding

Ans: B

17.Which of the following is illegal? (L1) (Page no 36)

A) int *ip;

B) string s, *sp = 0;

C) int i; double* dp = &i;

D) int *pi = 0;

Ans: C

18.Which member can never be accessed by inherited classes? (L1) (Page no 45)

A) Private member function

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B) Public member function

C) Protected member function

D) All can be accessed

Ans: A

19.Analyze the code and choose the correct Ans? (L4) (Page no 37)

int a=100, b=200;

int *p=&a, *q=&b;

p=q;

A) b is assigned to a

B) p now points to b

- C) a is assigned to b
- D) q now points to a

Ans: B

20. Mention the size_t integer type in C++ is? (L2) (Page no 52)

- A) Unsigned integer of at least 64 bits
- B) Signed integer of at least 16 bits
- C) Unsigned integer of at least 16 bits
- D) Signed integer of at least 64 bits

Ans: c

4 Marks

21. Abstraction and encapsulation are fundamental principles that underlie the object oriented

approach to software development. What can you say about the following two statements? (L2) (Page no 22)

I. Abstraction allows us to focus on what something does without considering the complexities of how it works.

II. Encapsulation allows us to consider complex ideas while ignoring irrelevant detail that would confuse us.

- A. Neither I nor II is correct
- B. Both I and II are correct
- C. Only II is correct
- D. Only I is correct

Ans: A

22. Class A

```
{ public :  
A(void)  
{cout << "Howzaht" ; }  
~ A(void)  
{cout << "whatizit " ; }  
};  
class B:A  
{ public:  
~ B (void)  
{  
cout << "WYSIWYG" ;
```

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}


```
cout << "YACC" ;
}
};
```

If the main function has the two statements (L2) (Page no 21)

```
B x ;
cout << " done" ;
the output will be
```

- A. Howzhat WYSIWYG YACC Whatizi t done
- B. Howzhat WYSIWYG done YACC Whatizit
- C. Yl>.CC Whatizit Howzhat WYSIWYG done
- D. none of the above

Ans: B

23. The following program (L2) (Page no 31)

```
class abc;
class def
{
int i1 ; // statement 1
protected: int i2; // statement 2
public: int i3; //statement 3
friend abc;
} ;
class abc
{
public:
void main(def A)
{
cout << (A.i1=3);
cout << (A.i2=4);
cout << (A.i3=5);
}
} ;
void main( )
{
def x1 ;
abc x2 ;
x2.mn(x1);
}
```

- A. will compile successfully if statement 1 is removed
- B. will compile successfully if statement 2 is removed
- C. will compile successfully if statement 3 is removed

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D. will run successfully and print 34 5

Ans D

24. What will be the output of the following code? (L2) (Page no 36)

Class A

```
{
    int i;
    public : A(int n)
    {
        i=n; cout<<<"inside constructor ";
    }
    ~A()
    {
        cout<<<"destroying "<<i;
    }
    void seti(int n)
    {
        i=n;
    }
    int geti()
    {
        return I;
    }
};

void t(A ob)
{
    cout<<<"something ";
}

int main()
{
    A a(1);
    t(a);
    cout<<<"this is i in main ";
    cout<<<a.geti();
}

A)inside constructor something destroying 2this is i in main destroying 1
B)inside constructor something this is i in main destroying 1
C)inside constructor something destroying 2this is i in main
D) something destroying 2this is i in main destroying 1
```

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Ans A

25. Match the following concepts and their best possible descriptions. (L2) (Page no 42)

.

(A) i-b, ii-d, iii-e, iv-f, v-g, vi-a (B) i-c, ii-a, iii-e, iv-d, v-h, vi-f

(C) i-c, ii-f, iii-h, iv-a, v-g, vi-d (D) i-b, ii-e, iii-c, iv-f, v-g, vi-s

Ans:A

26. Predict the output of following C++ program(L2) (page no 43)

```
#include<iostream>
using namespace std;
class Empty {};
int main()
{
    cout << sizeof(Empty);
    return 0;
}
```

A. A non-zero value

B. 0

C. Compiler Error

D. Runtime Error

Ans:A

27. Predict the output following program. (L2) (Page no 43)

```
#include<iostream>
using namespace std;
class Test
```

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```
{

static int x;
int *ptr;
int y;
};
int main()
{
```

```

Test t;
cout << sizeof(t) << " ";
cout << sizeof(Test *);
}

```

- A. 12 4
- B. 12 12
- C. 8 4
- D. 8 8

Ans:C

28. Output of following program? (L2) (page no 43)

```

#include<iostream>
using namespace std;
class Point {
public:
    Point() { cout << "Normal Constructor calledn"; }
    Point(const Point &t) { cout << "Copy constructor calledn"; }
};
int main()
{
    Point *t1, *t2;
    t1 = new Point();
    t2 = new Point(*t1);
    Point t3 = *t1;
    Point t4;
    t4 = t3;
    return 0;
}

```

A. Normal Constructor called
Normal Constructor called

Normal Constructor called

Copy Constructor called

Copy Constructor called

Normal Constructor called

Copy Constructor called

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B. Normal Constructor called

Copy Constructor called

Copy Constructor called

Normal Constructor called

Copy Constructor called

C.Normal Constructor called

Copy Constructor called

Copy Constructor called

Normal Constructor called

D.Normal Constructor called

Copy Constructor called

Copy Constructor called

Ans:C

29.Analyze the Output. (L4) (Page no 44)

```
#include<iostream>
```

```
using namespace std;
```

```
class Test
```

```
{
```

```
public:
```

```
Test();
```

```
};
```

```
Test::Test() {
```

```
cout << " Constructor Called. ";
```

```
}
```

```
void fun() {
```

```
static Test t1;
```

```
}
```

```
int main() {
```

```

cout << " Before fun() called. ";
fun();
fun();
cout << " After fun() called. ";
return 0;
}

```

A. Constructor Called. Before fun() called. After fun() called.

B. Before fun() called. Constructor Called. Constructor Called. After fun() called.

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C. Before fun() called. Constructor Called. After fun() called.

D. Constructor Called. Constructor Called. After fun() called. Before fun() called.

Ans:C

30. What is the output of following program? (L2) (page no 43)

```

#include <iostream>
using namespace std;

```

```

class Point
{
    int x, y;
public:
    Point(const Point &p) { x = p.x; y = p.y; }
    int getX() { return x; }
    int getY() { return y; }
};

```

```

int main()
{
    Point p1;
    Point p2 = p1;
    cout << "x = " << p2.getX() << " y = " << p2.getY();
    return 0;
}

```

a. x = garbage value y = garbage value

b. x = 0 y = 0

c. Compiler Error

d. Runtime

Ans: C

12Marks

31. Consider the following class definitions in a hypothetical Object-Oriented language that supports inheritance and uses dynamic binding. The language should not be assumed to be either Java or C++, though the syntax is similar. (L4) (Page no 62)

Class P

```
{  
    void f(int i)  
    {print(i);  
    }  
}
```

Class Q subclass of P

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```
{  
    void f(int i)  
    {  
        print(2*i);  
    }  
}
```

Now consider the following program fragment:

```
P x = new Q();  
Q y = new Q();  
P z = new Q();  
x.f(1); ((P)y).f(1); z.f(1);
```

Here ((P)y) denotes a typecast of y to P. The output produced by executing the above program fragment will be

- a.121
- b.211
- c.212
- d.222

Ans: D

32. Which of the following is true about the following program(L4) (Page no 72)

```
#include <iostream>
class Test
{
public:
    int i;
    void get();
};
void Test::get()
{
    std::cout << "Enter the value of i: ";
    std::cin >> i;
}
Test t; // Global object
int main()
{
    Test t; // local object
    t.get();
    std::cout << "value of i in local t: "<<t.i<<'\\n';
    ::t.get();
    std::cout << "value of i in global t: "<<::t.i<<'\\n';
    return 0;
}
```

a.Compiler Error: Cannot have two objects with same class name

b.Compiler Error in Line “::t.get();”

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c. Compiles and runs fine

d. Runtime Error

Ans:C

33. What will be the out of the following program? (L4) (page no 56)

```
#include<iostream.h>
class BixBase
{
protected:
    int x, y;
public:
    BixBase(int xx = 0, int yy = 0)
    {
```



```

x = xx;
y = yy;
}
void Show()
{
cout<< x * this->y << endl;
}
};
class BixDerived
{
private:
BixBase objBase;
public:
BixDerived(int xx, int yy) : objBase(xx, yy)
{
objBase.Show();
}
~BixDerived()
{ }
};
int main()
{
BixDerived objDev(10, 20);
return 0;
}
A.0 B.100 C.200 D.400

```

Ans C

34. Analyze the Output. (L4) (Page no 74)

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```

#include <iostream>
using namespace std;
class A
{
int id;
static int count;
public:
A() {

```

```

count++;
id = count;
cout << "constructor for id " << id << endl;
}
~ A() {
cout << "destructor for id " << id << endl;
}
};

```

```
int A::count = 0;
```

```
int main() {
A a[3];
return 0;
}

```

```

A.constructor for id 1
constructor for id 2
constructor for id 3
destructor for id 3
destructor for id 2
destructor for id 1
B.constructor for id 1
constructor for id 2
constructor for id 3
destructor for id 1
destructor for id 2
destructor for id 3
C.Compiler Dependent.
D.constructor for id 1
destructor for id 1

```

Ans:A

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35. Analyze the output. (L4) (Page no 53)

Class A

```

{
int i;
public : A(int n)
{

```

```

i=n; cout<<<"inside constructor ";
}
~A()
{
cout<<<"destroying "<<i;
}
void seti(int n)
{
i=n;
}
int geti()
{
return I;
}
};
void t(A ob)
{
cout<<<"something ";
}
int main()
{
A a(1);
t(a);
cout<<<"this is i in main ";
cout<<<a.geti();
}

```

- A. inside constructor something destroying 2this is i in main destroying 1
B. inside constructor something this is i in main destroying 1
C. inside constructor something destroying 2this is i in main
D. something destroying 2this is i in main destroying 1

Ans : A

36.Analyze the Output (L4) (Page no 68)

```

template <typename T>
void test(const T&x)
{
static int count = 0;
cout <<< "x = " << x <<< " count = " << count <<< endl;
++count;
}

```

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```
    return;
}

void main()
{
    test<int> (2);
    test<int>(2);
    test<double>(2.2);
}
```

A.x = 2 count = 0

x = 2.2 count = 0

x = 2.2 count = 0

B.x = 2 count = 0

x = 2 count = 0

x = 2.2 count = 0

C.x = 2 count = 0

x = 2 count = 1

x = 2.2 count = 0

D.x = 2 count = 0

x = 2 count = 1

x = 2.2 count = 2

Ans:C

UNIT -II

1 Mark

1. While overloading binary operators using member function, it requires ____ arguments.

[L1]Page no:328

- a. Zero
- b. One
- c. Two
- d. Three

Ans: b

2. Which of the followings are true about constructors?[L1]Page no:227

- 1. A class can have more than one constructor.
- 2. They can be inherited.
- 3. Their address can be referred.
- 4. Constructors cannot be declared in protected section of the class.
- 5. Constructors cannot return values.

- a. Only 1, 2, 4
- b. 1,2,4,5
- c. 1, 3, 5
- d. 1, 4, 5

Ans: d

3. Which of the following keyword is used to overload an operator?[L1]Page no:319

- a. overload
- b. operator
- c.friend

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d.override

Ans: b

4. What will happen if a class is not having any name?[L1]Page no:238

- a. it cannot have a destructor
- b. It cannot have a constructor.
- c. It is not allowed.
- d. Both A and B

Ans: d

5. Which inheritance type is used in the class given below? [L1] Page no:319

```
class A : public X, public Y
```

- a. Multilevel inheritance
- b. Multiple inheritance
- c. Hybrid inheritance
- d .Hierarchical Inheritance

Ans: b

6. Which of the following operators cannot be overloaded?[L1] Page no:319

- a. []
- b. ->
- c. ?:
- d. *

Ans:c

7. In which of the following a virtual call is resolved at the time of compilation?[L2] Page no:319

- a. From inside the destructor.
- b. From inside the constructor.
- c. From inside the main ().
- d. Both A and B.

Ans: d

8. Which of the following operator is overloaded for object cout?[L1] page no:319

- a. >>
- b. <<
- c. +
- d. =

Ans:b

9. Assume class TEST. Which of the following statements is/are responsible to invoke copy constructor? [L2]

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- a. TEST T2 (T1)
- b. TEST T4 = T1
- c. T2 = T1
- d. both a and b

Ans: d

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10. Which of the following is the perfect set of operators that can't be overloaded in C++? [L2] Page no:319

- a. +=, ?, ::, >>
- b. >>, <<, ?, *, sizeof()
- c. ::, ., .*, ?:
- d. ::, ->, *, new, delete

Ans: c

11. How many operators are supported by C++?

[L1] page no:319

- a. 30 operators
- b. 40 operators
- c. 45 operators
- d. 65 operator

Ans: c

12. A non-member function that is given access to all members of a class within it is declared, is called [L1] Page no:319

- a. Access function
- b. Friend function
- c. Operator functions
- d. None of them

Ans: b

13. Which of the following operators should be preferred to overload as a global function rather than a member method? [L1] Page no:319

- a. Postfix ++
- b. Comparison Operator
- c. Insertion Operator <<
- d. Prefix++

Ans: c

14. We can overload which of the following C++ operators. [L2] Page no:319

- a. Arithmetic operator (+, -, *, /)
- b. Class Member Access Operators (., .*)
- c. Size operator (sizeof)
- d. Conditional operator (?:)

Ans: a

15. Operator overloading is also called polymorphism.[L1] Page no:319

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- a. run time
- b. initial time
- c. Compile time
- d. Completion time

Ans: c

16. Operator overloading is done with the help of a special function called, which describes the special task of an operator. [L1] Page no:319

- a. overloading function
- b. special task function
- c. detail function
- d. operator function

Ans:d

17. Overload an operator by naming it a[L1] Page no:319

- a. variable
- b. built-in type
- c. function
- d. class.

Ans:c

18. Which of the function operator cannot be over loaded[L2] Page no:319

- a. <=
- b. ?:
- c. ==
- d. *

Ans:b

19. Kind of diagrams which are used to show interactions between series of messages are classified as[L1]

Page no:357

- a. activity diagrams
- b. state chart diagrams
- c. collaboration diagrams
- d. object lifeline diagrams

Ans:c

20. Dynamic aspects related to a system are shown with help of[L1] Page no:35

- a. sequence diagrams
- b. interaction diagrams
- c. deployment diagrams

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- d. use case diagrams

Ans:b

4 Marks

21. Locate which of these are true with respect to the message arrows?[L2] Page no:357

- a) The synchronous message arrow is used when a sending individual continues execution after sending the message
- b) The asynchronous message arrow is used when a sending individual suspends execution after sending the message
- c) The dashed arrow is used either to show the return of control from a synchronous message or to create a new entity
- d) All of the mentioned

Ans:c

22. Recognize the following statements and find which is correct ?[L2] Page no:371

- a)..Base class pointer cannot point to derived class.
- b).Derived class pointer cannot point to base class.
- c).Pointer to derived class cannot be created.
- d).Pointer to base class cannot be created.

Ans:b

23. Implement and find the output of the following C++ code?[L3] Page no:319

```
#include <iostream>
#include <string>
using namespace std;
```

```

class Box{
    int capacity
public:
    Box(){}
    Box(double capacity){
        this->capacity = capacity;
    }
    bool operator<(Box b){
        return b.capacity < this->capacity? true : false;
    }
};

```

```

int main(int argc, char const *argv[])
{
    Box b1(10);
    Box b2 = Box(14);
    if(b1 < b2){
        cout<<"B1's capacity is small";
    }
    else{
        cout<<"B2's capacity is small";
    }
    return 0;}

```

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- a) B1's capacity is small
- b) B2's capacity is small
- c) Error
- d) Segmentation fault

Ans:b

24.Solve the function prototype of the operator function which we need to define in this program so that the program has no errors?[L3] Page no:319

```

#include <iostream>
#include <string>
using namespace std;
class Box{
    int capacity;
public:
    Box(){}

```

```

Box(double capacity){
    this->capacity = capacity;
}
};
int main(int argc, char const *argv[])
{
    Box b1(10);
    Box b2 = Box(14);
    if(b1 == b2){
        cout<<"Equal";
    }
    else{
        cout<<"Not Equal";
    }
    return 0;
}
a) bool operator==( );
b) bool operator==(Box b){}
c) bool operator==(Box b);
d) Box operator==( );

```

Ans:c

25.Judge the operator to be overloaded in the following code to make the program error free?

[L5]Page no:319

```

#include <iostream>
#include <string>
using namespace std;
class Box{
    int capacity;
public:
    Box(){}
    Box(double capacity){
        this->capacity = capacity;
    }

```

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```

};
int main(int argc, char const *argv[])
{
    Box b1(10);

```

```

Box b2 = Box(14);
if(b1 == b2){
cout<<"Equal";
}
else{
c out<<"Not Equal";
}
return 0;
}

```

- a) +
 - b) ==
 - c) =
 - d) ()
- Ans:b

26. Define the syntax of overloading operator + for class A? [L1] Page no:319

- a) A operator+(argument_list){}
- b) A operator[+](argument_list){}
- c) int +(argument_list){}
- d) int [+](argument_list){}

Ans:a

27. State which of the following statements is NOT valid about operator overloading? [L1] Page no:319

- a) Only existing operators can be overloaded
- b) The overloaded operator must have at least one operand of its class type
- c) The overloaded operators follow the syntax rules of the original operator
- d) The overloaded operator must not have at least one operand of its class type.

Ans:d

28. Identify the three different types of message arrows? [L2] Page no:357

- a) Synchronous, asynchronous, asynchronous with instance creation
- b) Self, Multiplied, instance generator
- c) Synchronous, Asynchronous, synchronous with instance creation
- d) None of the mentioned

Ans:c

29. Consider the declarations? [L1] Page no:319

```

char a;
const char aa = 'h';
char *na;

```

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```
const char *naa;
```

Which of the following statements

Statement I: aa = a;

Statement II: na = &a;

Statement III: na = &aa;

is/are illegal?

a. Only I and II

b.

Only II and III

c.

Only I and III

d.

All the three statements are illegal

Ans : c

30) Relate the differences between constructors and destructors and find which one is correct ? [L4] Page

no:357

a.

Constructors can take arguments but destructors cannot.

b.

Constructors can be overloaded but destructors cannot be overloaded.

c.

Destructors can take arguments but constructors cannot.

d.

Both (a) and (b)

Ans:d

12 Marks

31. Investigate which property of Object Oriented Programming is exhibited by the code? [L6]

page no:334

```
#include< iostream>
```

```
class quest
```

```
{
```

```
public:
```

```
float add( float a, float b)
```

```
{
```

```
cout << "The sum is:";
```

```
return(a+b); } }
```

```
int add( int a, int b, int c)
```

```
{
```

```
cout << "The sum is:";
```

```
return(a+b+c);
```

```
}
```

```
int add( int a, int b=0)
{
    cout << "The sum is:";
```

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```
    return(a+b);
}
};
int main( )
```

```
{
    quest q
    q. add( 50.5, 48.2)
    q. add( 42, 56, 82)
    q. add( 23)
    return 0;
}
```

- a. Compile time polymorphism
- b. Run time polymorphism
- c. polymorphism
- d. virtual polymorphism

Ans:a

32.Implement and find the output of the following C++ code?

[L3]Page no:319

```
#include <iostream>
using namespace std;
class sample
{
public:
    int x, y;
    sample() {}
    sample(int, int);
    sample operator + (sample);
};
sample::sample (int a, int b)
{
    x = a;
```

```

y = b;
}
sample sample::operator+ (sample param)
{
sample temp;
temp.x = x + param.x;
temp.y = y + param.y;
return (temp);
}
int main ()
{
sample a (4,1);
sample b (3,2);
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```

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```

sample c;
c = a + b;
cout << c.x << "," << c.y;
return 0;
}
a) 5, 5
b) 7, 3
c) 3, 7
d) 3, 5
Ans:b

```

33.Judge the output of the following C++ code? [L5] Page no:319

```

#include <iostream>
using namespace std;
class Integer
{
int i;
public:
Integer(int ii) : i(ii) {}
const Integer
operator+(const Integer& rv) const
{
cout << "operator+" << endl;
return Integer(i + rv.i);
}
}

```

```

}
Integer&
operator+=(const Integer& rv)
{
cout << "operator+=" << endl;
i += rv.i;
return *this;
}
};
int main()
{
int i = 1, j = 2, k = 3;
k += i + j;
Integer ii(1), jj(2), kk(3);
kk += ii + jj;
}

```

a)

```

operator+
operator+=

```

b)

```

operator+=
operator+

```

c)

```

operator+
operator+

```

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26

d)

```

operator+
operator=

```

Ans:a

34. Identify the output of the following C++ code? [L2] Page no: 319

```

#include <iostream>
using namespace std;
class myclass
{
public:
int i;
myclass *operator->()
{return this;}
}

```



```
};
int main()
{
myclass ob;
ob->i = 8;
cout << ob.i << " " << ob->i;
return 0;
}
```

- a) 8 8
- b) 11 11
- c) 12 12
- d) 0 0

Ans:A

35.Solve the following C++ code? [L3] Page no:319

```
#include <iostream>
#include <string>
using namespace std;
class A
{
static int a;
public:
void show()
{
a++;
cout<<"a: "<<a<<endl;
}
};
```

```
int A::a = 5;
```

```
int main(int argc, char const *argv[])
{
A a;
return 0;
```

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27

```
}
```

- a)Error as a private member a is referenced outside the class
- b) Segmentation fault

c) No output

d) Program compiles successfully but gives run-time error

Ans:c

36.Execute the following C++ code?[L3] Page no:319

```
#include <iostream>
#include <string>
using namespace std;
class A
{
    static int a;

    public:
    void show()
    {
        a++;
        cout<<"a: "<<a<<endl;
    }
    void operator.()
    {
        cout<<"Objects are added\n";
    }
};

class B
{
    public:
};

int main(int argc, char const *argv[])
{
    A a1, a2;
    return 0;
}
```

a)Run-time Error

b) Runs perfectly

c) Segmentation fault

d) Compile-time error

Ans:d

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28

UNIT -III

1 Mark

1. How many basic types of inheritance are provided as OOP feature?[Page No: 371] [L1]

- a) 4
- b) 3
- c) 2
- d) 1

Ans: a

2. Which among the following best defines single level inheritance?[Page No: 404] [L1]

- a) A class inheriting a derived class
- b) A class inheriting a base class
- c) A class inheriting a nested class
- d) A class which gets inherited by 2 classes

Ans: b

3. Which programming language doesn't support multiple inheritances?[Page No: 404] [L1]

- a) C++ and Java
- b) C and C++
- c) Java and Small Talk
- d) Java

Ans: d

4. Which access type data gets derived as private member in derived class?[Page No: 376] [L1]

- a) Private
- b) Public
- c) Protected
- d) Protected and Private

Ans: a

5. How can you make the private members inheritable?[Page No: 396] [L1]

- a) By making their visibility mode as public only
- b) By making their visibility mode as protected only
- c) By making their visibility mode as private in derived class
- d) It can be done both by making the visibility mode public or protected

Ans: d

6. Which problem arises due to multiple inheritances, if hierarchical inheritance is used previously for its base classes?[Page No: 406] [L1]

- a) Diamond
- b) Circle
- c) Triangle
- d) Loop

Ans: a

7. How many classes should a program contain to implement the multiple inheritance?

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[Page No: 410] [L1]

- a) Only 1
- b) At least 1
- c) At least 3
- d) Exactly 3

Ans: c

8. Is it compulsory to have constructor for all the classes involved in multiple inheritance?

[Page No: 414] [L1]

- a) Yes, always
- b) Yes, only if no abstract class is involved
- c) No, only classes being used should have a constructor
- d) No, they must not contain constructors

Ans: b

9. Can the derived class be made abstract if multiple inheritances is used?[Page No: 407] [L1]

- a) No, because other classes must be abstract too
- b) Yes, if all the functions are implemented
- c) Yes, if all the methods are predefined
- d) No, since constructors won't be there

Ans: d

10. Name the function whose definition can be substituted at a place where its function call is made: ?

[Page No: 164] [L1]

- a) friends function
- b) inline function
- c) volatile function
- d) external function

Ans: b

11. Which keyword is used to declare the friend function?[Page No: 520] [L1]

- a) friend
- b) friend
- c) classfriend
- d) myfriend

Ans: b

12. Which of the given modifiers can be used to prevent Method overriding? [Page No: 371] [L1]

- a) Static
- b) Constant
- c) Sealed
- d) final

Ans: c

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13. Which of the following cannot be used to declare a class as a virtual?[Page No: 504] [L1]

- a) Methods
- b) Properties
- c) Events
- d) Fields

Ans: d

14. Can abstract class have main () function defined inside it?[Page No: 510] [L1]

- a) Yes, depending on return type of main()
- b) Yes, always
- c) No, main must not be defined inside abstract class

d) No, because main() is not abstract function

Ans: b

15. If there is an abstract method in a class then, _____[Page No: 511] [L1]

- a) Class must be abstract class
- b) Class may or may not be abstract class
- c) Class is generic
- d) Class must be public

Ans: a

16. Which of the following UML diagrams has a static view?[Page No: 160] [L1]

- a) Collaboration
- b) Use case
- c) State chart
- d) Activity

Ans: b

17. Which diagram in UML shows a complete or partial view of the structure of a modeled system at a specific time?[Page No: 162] [L1]

- a) Sequence Diagram
- b) Collaboration Diagram
- c) Class Diagram
- d) Object Diagram

Ans: d

18. Use of pointers or reference to an abstract class gives rise to which among the following feature?

[Page No: 512] [L1]

- a) Static Polymorphism
- b) Runtime polymorphism
- c) Compile time Polymorphism
- d) Polymorphism within methods

Ans: b

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19. Activity diagram, use case diagram, collaboration diagram and sequence diagram are considered

as types of ?[Page No: 186] [L1]

- a) non-behavioral diagrams
- b) nonstructural diagrams
- c) structural diagrams
- d) behavioral diagrams

Ans: d

20. Diagrams which are used to distribute files, libraries and tables across topology of hardware are

called?[Page No: 188] [L1]

- a) deployment diagrams
- b) use case diagrams
- c) sequence diagrams
- d) collaboration diagrams

Ans: d

4 Marks

21. Which among the following is correct for multiple inheritance? [Page No: 405] [L2]

- a) `class student{public: int marks;}; class stream{int total;}; class topper:public student, public stream{ };`
- b) `class student{int marks;}; class stream{ }; class topper: public student{ };`
- c) `class student{int marks;}; class stream:public student{ };`
- d) `class student{ }; class stream{ }; class topper{ };`

Ans: a

22. While inheriting a class, if no access mode is specified, then which among the following is true? (in C++) [Page No: 398] [L1]

- a) It gets inherited publicly by default
- b) It gets inherited protected by default
- c) It gets inherited privately by default
- d) It is not possible

Ans: c

23. What is the output of this program? [Page No: 520] [L3]

```
#include <iostream>
using namespace std;
class sample
{
private:
int a, b;
public:
void test()
```

```
{  
a = 100;  
b = 200;
```

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```
}  
friend int compute(sample e1);  
};  
int compute(sample e1)  
{  
    return int(e1.a + e1.b) - 5;  
}  
int main()  
{  
    sample e;  
    e.test();  
    cout << compute(e);  
    return 0;  
}
```

- a) 100
- b) 200
- c) 300
- d) 295

Ans: d

24. Point out the error (if any) in the code shown below: [Page No: 196] [L3]

```
#include <stdio.h>  
void inline func1(float b)  
{  
    printf ("%lf\n",b*2);  
}  
int main()  
{  
    inline func1(2.2);  
    return 0;  
}
```

- a) No error
- b) Error in statement: void inline func1(float b)
- c) Error in statement: printf(“%lf\n”,b*2);
- d) Error in statement: inline func1(2.2);

Ans: d

25. What is the output of the code shown below? [Page No: 522] [L2]

```
#include <stdio.h>
void inline func1(char b[10])
{
    printf ("%c\n",b[2]);
}
int main()
{
    func1("sanfoundry");
    return 0;
}
```

a) s

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b) n

c) a

d) error

Ans: b

26. What is the output of this program? [Page No: 518] [L3]

```
#include <iostream>
using namespace std;
class sample
{
private:
    int a, b;
public:
    void test()
    {
        a = 100;
        b = 200;
    }
    friend int compute(sample e1);
};
int compute(sample e1)
{
```

```

return int(e1.a + e1.b) - 5;
}
int main()
{
sample e;
e.test();
cout << compute(e);
return 0;
}

```

a) 100

b) 200

c) 300

d) 295

Ans: d

27. What is the output of this program? [[Page No: 395] L3]

```

#include <iostream>
using namespace std;

```

```

class A
{
public:
    virtual void fun() { cout << "A::fun() "; }
};

```

```

class B: public A
{
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```

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34

```

public:
    void fun() { cout << "B::fun() "; }
};

```

```

class C: public B
{
public:
    void fun() { cout << "C::fun() "; }
};

```

```

int main()

```

```

{
    B *bp = new C;
    bp->fun();
    return 0;
}

```

a) A::fun() b) B::fun() c) C::fun() d) A:B:C:fun()

Ans: c

28. What type of core-relationship is represented by the symbol in the figure below?
[Page No:195] [L1]

- a) Aggregation
- b) Dependency
- c) Generalization
- d) Association

Ans: a

29. Which core element of UML is being shown in the figure? [Page No: 280] [L1]

- a) Node
- b) Interface
- c) Class
- d) Component

Ans: d

30. If class A inherits class B and class C as “class A: public class B, public class C {
class body ;};”,

which class constructor will be called first? [Page No: 397] [L2]

- a) Class A
- b) Class B
- c) Class C
- d) All together

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Ans: b

12 Marks

31. (i) When the inheritance is private, the private methods in base class are _____ in the

derived class. [Page No: 396] [L2]

- a. inaccessible
- b. Accessible
- c. Protected
- d. Public

(ii) What will be the order of execution of base class constructors in the following method of

inheritance?

class a: public b, public c {...};

- a. b(); c(); a();
- b. c(); b(); a();
- c. a(); b(); c();
- d. b(); a(); c();

iii) What are the things are inherited from the base class?

- a. Constructor and its destructor
- b. Operator = () members
- c. Friends
- d. All of the above

Ans: (i)a (ii)a (iii)d

32. Consider the following code [Page No: 399] [L3]

```
class a1 {
public:
    ~a1() { cout << " a1" << endl; }
};
```

```
class a2 {
public:
    ~a2() { cout << " a2" << endl; }
};
```

```
class c: public a1, public a2 {
public:
    ~derived() { cout << " derived" << endl; }
};
```

```
int main()
{
    Derived d;
```

```
return 0;
```

```
}
```

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36

(i) The output of the above code is

a. a1

a2

derived

b. derived

a2

a1

c .derived

d. a2

(ii) Which design patterns benefit from the multiple inheritances?

a. Adapter and observer pattern

b.Code pattern

c. Glue pattern

d. None of the mentioned

(iii) What is the ability to group some lines of code that can be included in the program?

a) Specific task

b) program control

c) modularization

d) macros

Ans: (i) b (ii) a (iii) c

33.Consider the following code [Page No: 382] [L3]

```
include<iostream>
```

```
using namespace std;
```

```
class Base
```

```
{
```

```
public:
```

```
virtual void show() { cout<<" In Base \n"; }
```

```
};
```

```

class Derived: public Base
{
public:
    void show() { cout<<"In Derived \n"; }
};

```

```

int main(void)
{
    Base *bp = new Derived;
    bp->show();
}

```

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[Page 37]

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```

Base &br = *bp;
br.show();

return 0;
}

```

(i) The o/p of above code is

a) in base b) in base c) in derived d) in derived
 in base in derived in derived in base

(ii) Can static functions be virtual? Will the following program compile?

```

#include<iostream>
using namespace std;

```

```

class Test
{
public:
    virtual static void fun() { }
};

```

a) Yes, virtual b) Not a virtual c) static d) based on compiler

(iii) Which is used to create a pure virtual function?

a) \$
 b) =0

- c) &
- d)!

(iv) Which is also called as abstract class?

- a) Virtual function
- b) pure virtual function
- c) derived class
- d) base class

Ans: (i)c (ii) a (iii) b (iv) b

34.(i) The element of state chart diagrams which uses round corner boxes to represent situations of an

object is classified as [Page No: 220] [L2]

- a) Lifeline marker b) iteration marker c) transitions d) state

(ii) The relationship between two independent data table is classified as

- a) Structural relationship b) Non structural relationship c) identifying relationship
- d) Non identifying relationship

(iii) In state chart diagrams, the element which is shown with the help of double line filled circle with

pointing arrow is classified as

- a) Initial state b) Two degree state c) Final state d) Zero degree state

(iv) In state chart diagrams, the element which is shown with the help of solid circle with outgoing

arrow is classified as

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- a) Initial state b) Two degree state c) Final state d) Zero degree state

Ans: (i) d (ii) d (iii)c (iv) a

35. (i) The elements included in state chart diagrams are [Page No: 220] [L2]

- a) Transitions b) Condition marker c) lifeline marker d) iteration marker

(ii) Which of the following determines the state diagram?

- a) The UML notation for specifying finite automata is the state diagram
- b) In state diagrams, states are represented by rounded rectangles
- c) All of the mentioned
- d) None of the mentioned

(iii) Which of the statements state the name compartment?

- a) The first compartment is the name compartment
- b) It contains the state name; State names are optional and may be path names
- c) The name compartment can never be omitted
- d) The first compartment is the name compartment, It contains the state name; State names are optional and may be path names

Ans: (i)a (ii)c (iii)d

36. (i) If class Apple inherits class Banana and class Cat as “class Apple: public class Banana,

public class Cat { // class body ;}; ”, which class constructor will be called first?[Page No: 403] [L3]

- a) Class Apple
- b) Class Banana
- c) Class Cat
- d) All together

(ii) Why does diamond problem arise due to multiple inheritance?

- a) Methods with same name creates ambiguity and conflict
- b) Methods inherited from the super class may conflict
- c) Derived class gets overloaded with more than two class methods
- d) Derived class can't distinguish the owner class of any derived method

(iii) When multiple inheritances is used, which class object should be used in order to access all the

available members of parent and derived class?

- a) Derived class object
- b) Parent class objects
- c) Use Abstract derived class
- d) Derive a class from derived class

(iv) Which members can't be accessed in derived class in multiple inheritance?

- a) Private members of base
- b) Public members of base

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- c) Protected members of base
- d) All the members of base

Ans: (i) b(ii)a (iii)a (iv)a

UNIT-IV

1 MARK

1. The STL can be used as a standard approach for-----[L1, R5-726]

- a) Storing and sorting
- b) Storing and processing data
- c) data processing only
- d) storing only

Ans:b

2. Name the Container which uses both stack and queue.[L1, R5-728]

- a) storage
- b) linked list
- c) queuing
- d) Deque

Ans:d

3. Identify the characteristics of vector container.[L2, R5-728]

- a) Relocating, expandable array
- b) Fixed size
- c) Doubly linked list
- d) link vector

Ans:a

4. Associative container uses-----to access data.[L1, R5-729]

- a) queue
- b) Keys
- c) stack
- d) string

Ans:b

5. Class templates are generally used for-----[L1, R5-690]

- a) Data storage
- b) debug
- c) fixed data type
- d) storage

Ans:a

6. In UML, Templates are also called as-----[L1, R5-702]

- a) container

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- b) modified
- c) Parameterized
- d) generic

Ans:c

7. ____ specifies additional detail about UML element[L1, R5-703]

- a) Stereotype
- b) container
- c) associative container
- d) data processing

Ans:a

8.----- is visible only to its containing package and to its nested package.[L1, R1-163]

- a) protected
- b) public
- c) Private
- d) package

Ans:c

9. Notation is used to specify the required and provided interfaces of the components. The interfaces between the components are named as-----[L1, R1-172]

- a) Assembly connectors
- b) cooling controllers
- c) Environmental controller
- d) Plan analyst

Ans:a

10. List the 3 essential elements of a deployment diagram?[L1, R1-177]

- a) Artifacts, nodes and connections.
- b) stack, queue, deque
- c) memory, database, connections
- d) package, element, deployment

Ans:a

11. Activity, use case diagram, collaboration diagram and sequence diagram are categorized as [L4, R1-147]

- a) non-behavioral diagrams
- b) non structural diagrams
- c) structural diagrams
- d) Behavioral diagrams

Ans:d

12. Recognize which diagram is used to distribute files, libraries and tables across topology of hardware?

[L1, R1-171]

- a) Deployment diagrams
- b) use case diagrams
- c) sequence diagrams
- d) collaboration diagrams

Ans:a

13. List the essentials in package diagram [L1, R1-165]

- a) Package notation, element visibility, dependency relationship
- b) package notation, sequence, dependency relationship
- c) Dependency, element visibility
- d) package, deployment, sequence

Ans:a

14. Good packages are-----coupled and highly cohesive among the elements in package.

[L1, R1-167]

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- a) Tightly
- b) highly
- c) loosely
- d) semi

Ans:c

15. Identify the validity of template parameters? [L1, R5-682]

- a) inside that block only
- b) inside the class
- c) whole program
- d) inside the main class

Ans:a

16. Identify the core element of UML in the below figure?[L2, R1-163]

- a) Node
- b) Interface
- c) Class
- d) Component

Ans:d

17. Recognize the UML diagram shown below? [L2, R1-163]

- a) Component
- b) Deployment
- c) Use Case
- d) DFD

Ans:a

18. -----type of program can be included in try block? [L1, R5-705]

- a) static memory allocation
- b) Dynamic memory allocation
- c) const reference
- d) pointer

Ans:b

19. -----statement is used to catch all types of exceptions. [L1, R5-705]

- a) catch()
- b) catch(Test t)
- c) catch(...)
- d) none of the mentioned

Ans:c

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20. The ----- class name must be included in the class in which it is located. [L1, R5-709]

- a) try
- b) Exception
- c) catch
- d) template

Ans:b

4 Mark

21. From where does the template class derived? [L2, R5-702]

1. Regular non-templated C++ class
2. Templated class

3. A or B
4. None of the above
a) Only 1st is correct
b) Only 2nd is correct
c) Both 1st and 2nd are correct
d) Maybe 1st or 2nd is correct.

Ans:d

22. Explore the correct statement about string template?[L3, R5-683]

- a) It is used to replace a string.
b) It is used to replace a string with another string at runtime.
c) It is used to delete a string.
d) None of the above

Ans:b

23. Identify which among the following is not correct. [L2, R5-683]

- a) `template <class T> func(T x) {}`
`template <class T> func<T*>(T* x) {}`
b) `template <class T>`
`class myObject {};`
c) `template <class T>`
`class myObj { template <class R> memFunc() {} };`
d) All of the above are correct.

Ans:a

24. Examine whether templates are conceptually related to polymorphism? [L4, R5-682]

- a) Not Related
b) Only when the template types are objects
c) Yes, but compile-time polymorphism
d) Yes, but run-time polymorphism

Ans:c

25. Identify an invalid template declaration.[L2, R5-683]

- a) `template <int x> int func() {return x;}`
b) `template <double x> double func() {return x;}`
c) `template <typename x> void func(x t) {}`

Ans:b

26. Relate an option to restrict a function to throw certain exceptions?[L3, R5-705]

- a) Defining multiple try and catch block inside a function
b) Defining generic function within try block
c) Defining function with throw clause

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d) It is not possible in CPP to restrict a function

Ans:c

27. Select the ways to represent nodes in a deployment diagram? [L6, R1-171]

- a) Nodes instances are underlined identifiers of the form name:type
- b) The name may be left off, indicating an unnamed instance of the type
- c) The type may be left off, indicating a named instance with an unspecified type
- d) All of the mentioned

Ans:d

28. Examine: In component diagrams, building block which is represented with two rectangles laid on left

side is classified as [L4, R1-163]

- a) type of components
- b) interfaces
- c) dependency relationships
- d) State dependency

Ans:c

29. Choose the ways to represent nodes in a deployment diagram? [L6, R1-171]

- a) Nodes instances are underlined identifiers of the form name:type
- b) The name may be left off, indicating an unnamed instance of the type
- c) The type may be left off, indicating a named instance with an unspecified type
- d) All of the mentioned

Ans:d

30. Identify the ways to represent nodes in a deployment diagram? [L2, R1-171]

- a. Nodes instances are underlined identifiers of the form name:type
- b. The name may be left off, indicating an unnamed instance of the type
- c. The type may be left off, indicating a named instance with an unspecified type
- d. All of the mentioned

Ans:d

12 Mark

31. Predict the output of the following C++ code? [L5, R5-682]

```
#include <iostream>
using namespace std;
template <class T, int N>
class mysequence
{
    T memblock [N];
public:
    void setmember (int x, T value);
```

```

T getmember (int x);
};
template <class T, int N>
void mysequence<T,N> :: setmember (int x, T value)
{
    memblock[x] = value;
}
template <class T, int N>
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```

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```

T mysequence<T,N> :: getmember (int x)
{
    return memblock[x];
}
int main ()
{
    mysequence <int, 5> myints;
    mysequence <double, 5> myfloats;
    myints.setmember (0, 100);
    myfloats.setmember (3, 3.1416);
    cout << myints.getmember(0) << '\n';
    cout << myfloats.getmember(3) << '\n';
    return 0;
}

```

- a) 100
- b) 3.1416
- c) 100
- 3.1416
- d) 4.14

Ans:c

32. Identify the best-defined syntax for template function? [L4, R5-684]

1. template return_type Function_Name(Parameters)
2. template
return_type Function_Name(Parameters)
3. tmp return_type Function_Name(Parameter1, parameter 2)
4. template
Return_type(parameters)

- a) Only 1st syntax is correct
- b) Only 2nd syntax is correct

- c) Both 1st and 2nd syntaxes are correct
- d) None of the above

Ans:c

33. Predict the output of the following C++ code? [L5, R5-682]

```
#include <iostream>
using namespace std;
template <class T>
T max (T& a, T& b)
{
    return (a>b?a:b);
}
int main ()
{
    int i = 5, j = 6, k;
    long l = 10, m = 5, n;
    k = max(i, j);
    n = max(l, m);
    cout << k << endl;
    cout << n << endl;
    return 0;
}
```

- a) 6

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- b) 6

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- c) 5

10

- d) 5

Ans:b

34. Predict the output of this program? [L5, R5-682]

```
#include < iostream >
#include < string >
using namespace std;
template < typename T >
void print_mydata(T output)
{
    cout << output << endl;
```



```

}
int main()
{
double d = 5.5;
string s("Hello World");
print_mydata( d );
print_mydata( s );
return 0;
}

```

- a) 5.5
- b) 5.5 Hello World
- c) Hello World
- d) None of the above

Ans:b

35. Predict the output of this program? [L5, R5-705]

```

#include <iostream>
using namespace std;
int main()
{
int x = -1;
try
{
if (x < 0)
{
throw x;
}
else
{
cout<<x;
}
}
catch (int x )
{
cout << "Exception occurred: Thrown value is " << x << endl;
}
return 0;
}

```

- a) -1

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- b) 0
- c) Exception occurred: Thrown value is -1
- d) Error

Ans:c

36. Predict the output of this program? [L5, R5-705]

```
#include <iostream>
#include <exception>
using namespace std;
void myunexpected ()
{
    cout << "unexpected called\n";
    throw 0;
}
void myfunction () throw (int)
{
    throw 'x';
}
int main ()
{
    set_unexpected (myunexpected);
    try
    {
        myfunction();
    }
    catch (int)
    {
        cout << "caught int\n";
    }
    catch (...)
    {
        cout << "caught other exception\n";
    }
    return 0;
}
```

- a) caught other exception
- b) caught int
- c) unexpected called
- d) both caught int & unexpected called

Ans:d

UNIT V

1 Mark

1. What kind of library is Standard Template Library? [Page No: 620] [L1]

- a) Polymorphic
- b) Generic
- c) Both Polymorphic & Generic
- d) None of the mentioned

Ans:b

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2. To what type of object does the container can be instantiated? [Page No: 650] [L1]

- a) int
- b) float
- c) double
- d) any type of object

Ans:d

3. What type of class template is list? [Page No: 679] [L1]

- a) Class-based
- b) Node-based
- c) Method-based
- d) None of the mentioned

Ans:b

4. What type of access does deque and vector provide? [Page No: 675] [L1]

- a) Linear access
- b) Parallel access
- c) Random access
- d) None of the mentioned

Ans:c

5. Where does the vector add the item? [Page No: 685] [L1]

- a) End
- b) Insert
- c) Middle
- d) None of the mentioned

Ans:a

6. Which are not full container classes in C++?[Page No: 655] [L1]

- a) Sequence container
- b) Associative container
- c) Container adaptor
- d) None of the mentioned

Ans:c

7. What is the lifetime of the element in container? [Page No: 660] [L1]

- a) Whole program
- b) Outside the block

- c) Everywhere
- d) Only on that container

Ans:d

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8. Which operator is used to insert the data into file? [Page No: 705] [L1]

- a) >>
- b) <<
- c) <
- d) None of the mentioned

Ans:b

9. Which function is used to position back from the end of file object? [Page No: 725] [L1]

- a) seekg
- b) seekp
- c) both seekg&seekp
- d) none of the mentioned

Ans:a

10. How many objects are used for input and output to a string? [Page No: 712] [L1]

- a) 1
- b) 2
- c) 3
- d) 4

Ans:c

11. Which is used to handle the exceptions in c++?[Page No: 745] [L1]

- a) catch handler
- b) handler
- c) exception handler
- d) none of the mentioned

Ans: c

12. Which type of program is recommended to include in try block? [Page No: 715] [L1]

- a) static memory allocation
- b) dynamic memory allocation
- c) const reference
- d) pointer

Ans:b

13. Which statement is used to catch all types of exceptions? [Page No: 706] [L1]

- a) catch()
- b) catch(Test t)
- c) catch(...)
- d) none of the mentioned

Ans:c

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14. How to handle error in the destructor? [Page No: 710] [L1]

- a) throwing
- b) terminate
- c) both throwing & terminate
- d) none of the mentioned

Ans:b

15. What kind of exceptions are available in c++?[Page No: 725] [L1]

- a) handled
- b) unhandled
- c) static
- d) dynamic

Ans:b

16. What do associate containers implement? [Page No: 670] [L1]

- a) Arrays
- b) Associative arrays
- c) Functional Arrays
- d) Static arrays

Ans: b

17. By using which of the following the elements in the associate container can be efficiently accessed? [Page

No: 656] [L1]

- a) Key
- b) Position
- c) Both Key & Position
- d) Value

Ans: a

18. How many items are presented in the associate container? [Page No: 658] [L1]

- a) 2
- b) 3
- c) 4
- d) 5

Ans: c

19. What are the containers? [Page No: 658] [L1]

- a) Containers store objects and data
- b) Containers stores all the algorithms
- c) Containers contain overloaded functions
- d) Containers contain set of Iterators

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Ans: a

20. In how many categories, containers are divided? [Page No: 660] [L1]

- a) 1
- b) 2
- c) 3
- d) 4

Ans:d

4 Marks

21.What are the Sequence Containers? [Page No: 675] [L2]

- a) Containers that implements data structures which can be accessed sequentially
- b) Containers that implements sorted data structures for fast search in $O(\log n)$
- c) Containers that implements unsorted(hashd) data structures for quick search in $O(1)$
- d) Containers that implements data structures which can be accessed non-sequentially

Ans:a

22.How many Sequence Containers are provided by C++?[Page No: 671] [L2]

- a) 2
- b) 3
- c) 4
- d) 5

Ans:d

23.What are the Associative Containers? [Page No: 667] [L2]

- a) Containers that implements data structures which can be accessed sequentially
- b) Containers that implements sorted data structures for fast search in $O(\log n)$
- c) Containers that implements unsorted(hashd) data structures for quick search in $O(1)$
- d) Containers that implements data structures which can be accessed non-sequentially

Ans:b

24. What are Container Adaptors? [Page No: 671] [L2]

- a) Containers that implements data structures which can be accessed sequentially
- b) Containers that implements sorted data structures for fast search in $O(\log n)$
- c) Containers that implements unsorted(hashd) data structures for quick search in $O(1)$
- d) Containers that provide a different interface for sequential containers

Ans:d

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25.What are Iterators? [Page No: 740] [L2]

- a) Iterators are used to iterate over C-like arrays
- b) Iterators are used to iterate over pointers
- c) Iterators are used to point memory addresses of STL containers
- d) Iterators are used to iterate over functions

Ans:c

26. How many types of Iterators are provided by C++?[Page No: 745][L2]

- a) 2
- b) 3
- c) 4
- d) 5

Ans:d

27. Which of the following statements are correct? [Page No: 654] [L2]

- a) It is not possible to combine two or more file opening mode in open() method.
- b) It is possible to combine two or more file opening mode in open() method.
- c) ios::in and ios::out are input and output file opening mode respectively.

Ans:a

28. Which function is used to reposition the file pointer? [Page No: 709] [L2]

- a) moveg()
- b) seekg()
- c) changep()
- d) go_p()

Ans:b

29. Where is a file temporarily stored before read or write operation in C language.? [Page No: 723] [L2]

- A) Notepad
- B) RAM
- C) Hard disk
- D) Buffer

Ans:d

30. What is the use of ios::trunc mode? [Page No: 725] [L2]

- a) To open a file in input mode
- b) To open a file in output mode

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- c) To truncate an existing file to half
- d) To truncate an existing file to zero Ans:d

12 Marks

31.Merge sort[Page No: 780] [L2]

i) Merge sort uses which of the following technique to implement sorting?

- a) backtracking
- b) greedy algorithm
- c) divide and conquer
- d) dynamic programming

ii) What is the worst case time complexity of merge sort?

- a) $O(n \log n)$
- b) $O(n^2)$
- c) $O(n^2 \log n)$
- d) $O(n \log n^2)$

iii) Which of the following method is used for sorting in merge sort?

- a) merging
- b) partitioning
- c) selection
- d) exchanging

iv) Which of the following is not a variant of merge sort?

- a) in-place merge sort
- b) bottom up merge sort
- c) top down merge sort
- d) linear merge sort

Answer

i)c

ii)a

iii)a

iv)d

32. File handling [Page No: 726] [L2]

i) Which header file is required to use file I/O operations?

- a) `<ifstream>`
- b) `<ostream>`

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- c) `<fstream>`
- d) `<iostream>`

ii) Which of the following is used to create an output stream?

- a) `ofstream`
- b) `ifstream`
- c) `iostream`
- d) `fsstream`

iii) Which of the following is used to create a stream that performs both input and output operations?

- a) ofstream
 - b) ifstream
 - c) iostream
 - d) fstream
- iv) Which of the following is not used as a file opening mode?

- a) ios::trunc
- b) ios::binary
- c) ios::in
- d) ios::ate

Answer

- i)c
- ii)a
- iii)d
- iv)a

33. What will be the output of the following C++ code? [Page No: 715] [L3]

```
i) #include <iostream>
using namespace std;
int main ()
{
    char first, second;
    cout << "Enter a word: ";
    first = cin.get();
    cin.sync();
```

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```
second = cin.get();
cout << first << endl;
cout << second << endl;
return 0;
}
```

- a)first
 - b)second
 - c)returns first 2 letter or number from the entered word
 - d) third
- ii) How many objects are used for input and output to a string?
- a) 1
 - b) 2
 - c) 3
 - d) 4
- iii) Which member function is used to determine whether the stream object is currently

associated with a
file?

- a) is_open
- b) buf
- c) string
- d) is_out

iv) Which function is used to position back from the end of file object?

- a) seekg
- b) seekp
- c) both seekg & seekp
- d) seekf

Answer

- i)c
- ii)c
- iii)a
- iv)a

34. Iterators[Page No: 749] [L2]

i) How many categories of iterators are there in c++?

- a) 2
- b) 4

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- c) 5
- d) 3

ii) What are Iterators?

- a) STL component used to point a memory address of a container
- b) STL component used for vectors
- c) STL component used to call functions efficiently
- d) STL component used to define template classes

iii) Which function is used increment the iterator by a particular value?

- a) next()
- b) advance()
- c) prev()
- d) move()

iv) Pick the correct statement.

- a) Input iterator moves sequentially forward
- b) Input iterator moves sequentially backward
- c) Input iterator moves in both direction
- d) Input iterator moves sequentially downwards

Answer

- i)c
- ii)a
- iii)b
- iv)a

35. Stack[Page No: 734] [L2]

i) Consider the usual algorithm for determining whether a sequence of parentheses is balanced.

The maximum number of parentheses that appear on the stack AT ANY ONE TIME when the algorithm analyzes:

((()((()((() are:

- a) 1
- b) 2
- c) 3
- d) 4 or more

ii) Process of removing an element from stack is called _____

- a) Create
- b) Push
- c) Evaluation
- d) Pop

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iii) Pushing an element into stack already having five elements and stack size of 5, then stack becomes

- a) Overflow
- b) Crash
- c) Underflow
- d) User flow

iv) Which of the following applications may use a stack?

- a) A parentheses balancing program
- b) Tracking of local variables at run time
- c) Compiler Syntax Analyzer
- d) Data Transfer between two asynchronous process

Answer

- i)c
- ii)d
- iii)a
- iv)d

36. Associative Container[Page No: 662] [L2]

i) What do associate containers implement?

- a) Arrays

- b) Associative arrays
 - c) Functional Arrays
 - d) Static arrays
- ii) By using which of the following the elements in the associate container can be efficiently accessed?
- a) Key
 - b) Position
 - c) Both Key & Position
 - d) Value
- iii) How many items are presented in the associate container?
- a) 2
 - b) 3
 - c) 4
 - d) 5
- iv) How many instances are allowed by map and set while inserting an element into container?
- a) 1
 - b) 2
 - c) 3
 - d) Multiple

Answer i)b ii)a iii)c iv)a

Downloaded by Vibhu Kaushik (vibhuk15@gmail.com)

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===== SOURCE_FILE: sem 2/Object Oriented Design And Programming/PYQs/More Questions.pdf

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CATEGORY: PYQs

STATUS: ok

[Page 1]

OOPS VIVA QUESTIONS

UNIT-1

1. What is OOP?

Answer:- Object-Oriented Programming is a programming paradigm / model that organizes software design around data or objects rather than functions

2. What is the difference between OOPS and POPS.

Answer:- POPS follows top down approach while OOPS follows bottom up approach.

In POP program is divided into smaller parts called function, while in oops program is divided

into smaller parts called objects.

3. What are the features of OOPS?

Answer:- Following are the 8 features of OOPS:-

I.) Objects:- An object is defined as an entity that contain data and its related function.

II.) Class:- A class is defined as a collection of objects with same type of data and functions.

III.) Data abstraction:- Abstraction means displaying only essential information and hiding the details. Data abstraction refers to providing only essential information about the data to the outside world, hiding the background details or implementation.

IV.) Data encapsulation:- In Object Oriented Programming, Encapsulation is defined as binding together the data and the functions that manipulates them.

V.) Inheritance:- It is defined as sharing of attributes and functions among classes based on a hierarchical or sequential relationship.

VI.) Polymorphism:- A function is said to be polymorphic if it is applied to many different classes with different operation.

VII.) Dynamic Binding:- Binding is defined as the connection between the function call and its

corresponding program code to be executed. In dynamic binding, the code to be executed in response to function call is decided at runtime.

VIII.) Message passing:- It is process of locating and executing a function in response to a message.

4. Name some benefits of OOP.

Answer:- 1. Data hiding is possible. 2. Reusability is possible. 3. Overloading is possible

5. What is method?

Answer:- Methods are functions that belongs to the class.

6. What is data types?

Answer:- A data type is used to indicate the type of data value stored in a variable.

7. What is primitive data type?

Answer:- These data types are built-in or predefined data types and can be used directly by the

user to declare variables. Example:- Integer, float, character, Boolean, etc

[Page 2]

8. What is derived data types?

Answer:- The data-types that are derived from the primitive or built-in data types.

Example:-

array, pointer, etc

9. What are user-defined data types?

Answer:- The data types defined by the user are known as the user-defined data types.

Example:- Structure, union, class, typedef,etc

10. What is variable? Answer:- A named memory location is called variable.

11. What is constants? Answer:- Constants are identifiers whose value does not change.

12. What is the type of constants in c++?

Answer:- Integer type, Floating point type, Character type, String type.

13. What is pointer?

Answer:- A pointer is a variable used to store the address of a memory cell.

14. What is generic pointer?

Answer:- Pointers of type void are known as generic pointers.

15. What is references?

Answer:- A special type of variable is called as enumerated or reference variable is used to provide alias (alternative name) for previously defined variable.

16. What is typecasting?

Answer:- Type conversion or typecasting of variables refers to changing a variable of one data type into another.

17. What is the use of scope resolution operator?

Answer:- In variables:- This is used to access global variables if same variables are declared as local and global.

In methods:- To define a member function outside the class definition we have to use the scope

resolution :: operator along with class name and function name.

18. What is access specifiers?

Answer:- Access specifier or access modifiers are the labels that specify type of access given to

members of a class. These are used for data hiding. These are also called as visibility modes.

19. Type of access specifier:- private public protected

20. What is argument?

Answer:- An argument is referred to the values that are passed within a function when the function is called. They are also called Actual Parameters

21. What is parameter?

Answer:- The values which are defined at the time of the function prototype or definition of the

function are called as parameters. They are also called Formal Parameters.

[Page 3]

22. What is UML?

Answer:- The Unified Modeling Language (UML) is the primary modeling language used to analyze, specify, and design software systems. UML diagrams can be classified into two groups:

structure diagrams and behavior diagrams.

23. What is structure diagrams?

Answer:- These diagrams are used to show the static structure of elements in the system.

24. What does structure diagram include?

Answer:- The UML structure diagrams include

■ Package diagram ■ Class diagram ■ Component diagram

■ Deployment diagram ■ Object diagram ■ Composite structure diagram

25. What is behavior diagrams?

Answer:- Those diagram in which the dynamic behavioral semantics of a problem or its implementation is depicted

26. What does UML behavior diagram includes?

Answer:- The UML Behavior diagrams include

■ Use case diagram ■ Activity diagram ■ State machine diagram

■ Interaction diagrams ■ Sequence diagram , etc

27. What is Package diagram?

Answer:- Package diagrams are structural diagrams used to show the organization and arrangement of various model elements in the form of packages.

28. What is class diagram?

Answer:- A class diagram is a view of the static structure of a system. It contains Packages, classes, interfaces, and relationships.

29. What is component diagram?

Answer:- Component diagrams are used in modeling the physical aspects of object-oriented systems that are used for visualizing, specifying, and documenting component-based systems.

30. What is Deployment diagram?

Answer:- A deployment diagram is a UML diagram type that shows the execution architecture of a system, including nodes such as hardware or software execution environments, and the middleware connecting them. Deployment diagrams are typically used to visualize the physical hardware and software of a system.

31. What is use-case diagram?

Answer:- A UML use case diagram is the primary form of system/software requirements for a new software program underdeveloped. Use cases specify the expected behavior (what), and not the exact method of making it happen (how).

[Page 4]

32. What is Activity diagram?

Answer:- An activity diagram is a behavioral diagram that portrays the control flow from a start point to a finish point showing the various decision paths that exist while the activity is being executed.

33. What is implementation Diagram?

Answer:- It shows the implementation phase of systems development, such as the source code structure and the run-time implementation structure.

34. What are the types of implementation diagram?

Answer:- There are two types of implementation diagrams:

1. component diagrams
2. deployment diagrams

35. What is component diagram?

Answer:- Component diagrams are used in modeling the physical aspects of object-oriented systems.

36. What is deployment diagram?

Answer:- Deployment diagrams are used to visualize the topology of the physical components of a system, where the software components are deployed.

1. What is constructor?

Answer:- A constructor is a special type of member function of a class which initializes objects of a class.

2. What is destructor?

Answer:- A destructor, is used to destroy the objects that have been created by a constructor.

Like a constructor, the destructor is a member function whose name is the same as the class name but is preceded by a tilde. Eg: ~item() { }

3. What are the types of constructor?

Answer:- Constructors are of three types

- Default Constructor
- Parameterized Constructor
- Copy Constructor Object

4. What is overloading?

Answer:- It allows multiple definitions of a function with the same name, but different signatures

[Page 5]

5. What is default constructor?

Answer:- Default constructor is the constructor which doesn't take any argument. it has no parameters.

6. What is parameterized constructor?

Answer:- These are the constructors with parameter. Using this Constructor you can provide different values to data members of different objects, by passing the appropriate values as argument.

7. What is Copy constructor?

Answer:- These are special type of Constructors which takes an object as argument, and is used

to copy values of data members of one object into other object. It creates a new object, which is

exact copy of the existing copy, hence it is called copy constructor.

8. Features of Polymorphism?

Answer:- Polymorphism means the ability to take more than one form. An operation have different behaviour in different instances.

9. What is constructor overloading/ multiple constructor?

Answer:- It means having more constructor with the same name as class name.

10. What are the types of polymorphism?

Answer:- C++ supports 2 types of polymorphism, 1. Compile time Polymorphism 2. Run time Polymorphism.

11. What is compile time polymorphism?

Answer:- The overloaded functions are invoked by matching the type and number of arguments. This information is available at the compile time and, therefore, compiler selects the appropriate

function at the compile time.

12. What is run time polymorphism?

Answer:- Run time polymorphism is achieved when the object's method is invoked at the run time instead of compile time.

13. What is function overloading?

Answer:- Function overloading is a feature of OOPs where two or more functions can have the same name but different parameters.

14. What is operator overloading?

Answer:- Operator overloading is a compile-time polymorphism in which the operator is overloaded to provide the special meaning to the user-defined data type. It use operator keyword.

[Page 6]

15. Operator that cannot be overloaded are as follows:

1. Scope operator (::)
2. Sizeof o member selector(.)
3. member pointer selector(*)
4. ternary operator(?:)

16. What are the types of operator overloading?

Answer:- Operator overloading types are

- Unary operator loading :- An unary operator means, an operator which works on single operand.
- Binary operator overloading :- An unary operator means, an operator which works on two operand.
- Operator Overloading using a friend function

UNIT- 3

1. What is inheritance?

Answer:- The capability of a class to derive properties and characteristics from another class is called Inheritance.

2. What are the types of inheritance?

Answer:-

- Single Inheritance: one derived class inherits from one base class.
- Multiple Inheritance: one derived class inherits from multiple base class(es)
- Multilevel Inheritance: wherein subclass acts as a base class for other classes. i.e a derived class is created from another derived class.
- Hierarchical Inheritance: wherein multiple subclasses inherited from one base class
- Hybrid (Virtual) Inheritance: Hybrid Inheritance is implemented by combining more than one type of inheritance. reflects any legal combination of other four types of

inheritance

3. What is base class/ Super class?

Answer:- The class whose properties are inherited by sub class is called Base Class or Super class.

Also known as parent class.

4. What is derived class/ Sub class?

Answer:- The class that inherits properties from another class is called Sub class or Derived Class.

Also know as child class.

[Page 7]

5. What are the various mode of inheritance?

Answer:- Following are the different mode of inheritance:-

1. Public mode
2. Private mode
3. Protected mode

6. What is inline function?

Answer:- Inline function is a function that is expanded in line when it is called. When the inline

function is called whole code of the inline function gets inserted or substituted at the point of

inline function call.

7. What is friend function?

Answer:- A friend function is a function that is declared outside a class, but is capable of accessing the private and protected members of class.

8. What is the syntax of friend function?

Answer:- friend ret_type func_name(arguments);

9. What is friend class?

Answer:- A friend function is a function that is declared outside a class, but is capable of accessing the private and protected members of class.

10. What is function/method overriding?

Answer:- If derived class defines same function as defined in its base class, it is known as function overriding in C++.

11. What is virtual function?

Answer:- A Virtual function is a member function in the base class whose definition is redefined

in derived classes.

12. Which keyword we use to make function virtual? Answer->> friend keyword.

13. What is Pure Virtual Function?

[Page 8]

Answer:- A pure virtual function (or abstract function) in C++ is a virtual function for

which we

can have implementation, But we must override that function in the derived class, otherwise the

derived class will also become abstract class.

14. What is an abstract class?

Answer:- A class containing pure virtual functions cannot be used to declare any objects of its

own. Such classes are called abstract base classes. OR Abstract Class is a class which contains

atleast one pure virtual function in it.

UNIT- 4

1. What is a template?

Answer:- A template is a simple and yet very powerful tool in C++. The simple idea is to pass data

type as a parameter so that we don't need to write the same code for different data types.

2. What are the types of templates?

Answer:- 1. Function template 2. Class template

3. Syntax of Templates:- `template <typename T>`

4. What is exception?

Answer:- Exceptions are run-time anomalies or abnormal conditions that a program encounters during its execution. Like we need 3 inputs and we get only 2 input, then there will be exception.

5. Name the keywords used for exception handling in c++.

Answer:- Try, catch, throw

6. What try keyword do?

Answer:- The try statement allows you to define a block of code to be tested for errors while it is

being executed.

7. What throw keyword do?

Answer:- The throw keyword throws an exception when a problem is detected, which lets us create a custom error.

8. What catch do?

Answer:- The catch statement allows you to define a block of code to be executed, if an error

occurs in the try blocks.

9. Which syntax we use when we don't know throw type?

Answer:- If we do not know the throw type used in the try block, you can use the "three dots"

syntax (...) inside the catch block, which will handle any type of exception.

10. What is interface?

Answer:- Interfaces in C++ are nothing but a way to describe the behavior of a class without any

commitment to any specific implementation of that class.

UNIT- 5

1. What is STL?

Answer:- The Standard Template Library (STL) is a set of C++ template classes to provide common programming data structures and functions such as lists, stacks, arrays, etc.

2. What are the components of STL?

Answer:- STL has four components:- 1. Algorithms 2. Containers 3. Functions 4. Iterators

3. What is algorithms?

Answer:- Algorithms in the STL are procedures that are applied to containers to process their data.

4. What are the types of algorithm in c++ stl?

Answer:- following are 5 types of algorithm in stl:-

1. Retrieve or Non-mutating Algorithms
2. Mutating Algorithms
3. Sorting Algorithms
4. Set Algorithms
5. Relational Algorithm

5. What is containers?

Answer:- Containers or container classes store objects and data.

6. What is iterators?

Answer:- Iterators are a generalization of the concept of pointers, they point to elements in a container

7. What is functions?

Answer:- The STL includes classes that overload the function call operator. Instances of such classes are called function objects or functors.

8. What are the types of iterators?

Answer:- there are 5 types of iterators:-

- i.) Input iterator:
- ii.) Output iterator
- iii.) Forward iterator
- iv.) Bi-directional iterator
- v.) Random access iterator

9. What are the type of containers?

Answer:- following are the types of containers:-

1. Sequence containers:- Arrays, list, deque, vector
2. Associative containers:- set, map, multiset, multimap
3. Derived containers:- stack, queue, priority_queue

[Page 10]

10. Commonly used containers:-

11. What is file?

Answer:- A file is a collection of information, usually stored on a computer's disk. Information

can be saved to files and then later reused.

12. What is stream?

Answer:- A transfer of information in the form of a sequence of bytes

13. Various types of streams:-

[Page 11]

Some list function:-

push_front(element) method is used to push elements into a list from the front.

push_back(element) method is used to push elements into a list from the back.

insert(iterator, element) : inserts element in the list before the position pointed by the iterator.

Empty() This method returns true if the list is empty else returns

Size() :- This method can be used to find the number of elements present in the list.

Reverse() :- This method can be used to reverse a list completely.

Sort() :- sort method sorts the given list

- The functions associated with stack are:

empty() - Returns whether the stack is empty

size() - Returns the size of the stack

top() - Returns a reference to the top most element of the stack

push(g) - Adds the element 'g' at the top of the stack

pop() - Deletes the top most element of the stack

SOME ALGORITHM:-

The find() algorithm looks for an element matching val between start and end

InputIterator find (InputIterator first, InputIterator last, const T& val);

count() returns the number of elements in the given range that are equal to given value.

count(first ,last ,value) :

The search() algorithm used to perform searches for a given sequence in a given range

search(first1 ,last1 ,first2 ,last2)

Sort() algorithm:- This function of the STL, sorts the contents of the given range

sort(start_iterator, end_iterator)

merge() :- Combines the elements in the sorted ranges [first1,last1) and [first2,last2),

into a

new range beginning at result with all its elements

```
OutputIterator merge (InputIterator1 first1, InputIterator1 last1, InputIterator2 first2,
InputIterator2 last2, OutputIterator result);
```

The for_each() algorithm allows you to do something to every item in a container

```
Function for_each (InputIterator first, InputIterator last, Function fn);
```

The transform() algorithm does something to every item in a container, and places the resulting values in a different container (or the same one).

```
OutputIterator transform (InputIterator first1, InputIterator last1, OutputIterator result,
UnaryOperation op);
```

NOTE:- THERE ARE LOTS AND LOTS OF FUNCTIONS FOR EACH CONTAINER TYPE IN STL(because it's a huge topic by itself) . BUT I DON'T THINK THAT THESE FUNCTIONS TYPE OF QUESTION CAN BE ASKED IN VIVA.

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Note: () (ii)

M.Tech. (Integrated) DEGREE EXAMINATION, DECEMBER 2022 Second and Third Semester

Part -A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed

over to hall invigilator at the end of 40h minute.

21CSC101T -OBJECT ORIENTED DESIGN AND PROGRAMMING (For the candidates admitted from the academic year 202/-2022)

Part -B should be answercd in answer booklet.

Time: 3 Hours

1. What happens when we try to compile class definition in following code? class Birds {};
class Peacock : protected Birds }; (A) It will not compile because class body of Birds is not defined

PART -A (20 x 1 = 20Marks) Answer ALL Questions

(B) It will not compile because class body of peacock is not defined (C) It will not compile because a class cannot be protectedly inherited from other class (D) It will compile successfully

2. Which of the following statement is correct regarding destructor of base class?

Page 1 of6

(A) Destructor of base class should (B) Destructor of base class should always be virtual always be static (C) Destructor of base class should (D) Destructor of base class should always be private not be virtual 3. Which of the following access specifiers is used in a class definition by default? (A) Protected (C) Public 4. Which of the following statements

is correct about program given below. `class Bix {public : static void myfunction (); }; int main () { void (*ptr) ) = & Bix :Myfunction; return 0; }` (A) Program reports an error as (B) pointer to member function

(B) Private (D) Friend

cannot be defined outside definition of class

member function cannot be (C) Reports an error as pointer to (D) The program reports linker defined without object.

Reports an error as pointer to static member function cannot be defined error

Max. Marks: 75

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5. Which among the following can be heuristic for use case diagram? (A) Product can be made actor

0. An operation can be described as?

(C) Name use case with verb (D) Product can be use case phrases

(A) Object behavior (C) actions 7. Find the output of the following code int main (01

'age 2 of 6

`cube ej class cube public: int side; cube() {side 10:};`

`cout << c.side:}` (A) 11 (C) 3

(A) Default constructor (C) Constructors

are special type of constructors which takes an object as argument, and its used to copy values of data members of one object into another object

(C) Defining more than one constructor in single class with different signature

(B) Never name actors with noun

be over loaded.

9, Which among the following best describes constructor overloading?

(A) =, 0, [], ' ,

phrases

(C) ?, =, (), + +

used.

(B) Class behavior

(A) Defining one constructor in (B) Defining more than One each class of a program constructor in single class

(D) Object and class behavior

(A) Activity (C) ClaSS

(B) 2 (D) 10

10. Using friend operator function, following perfect set of operators may not

(A) class student public int marks; s; class stream int total; }; class topper : public student, public stream }; (C) class student (int marks;): class stream : public student

(B) Parameterized constructor (D) Copy constructors

11. To model life time of a reactive system

(D) Defining destructor with each constructor

(B) <<, ==), >> (D) +, -, --, ++

(B) Use case UML diagrams can be

12. Which among the following is correct for multiple inheritance? (D) Statechart

(B) class student int marks;: class stream {}; class topper: public student : (D) class student } class stream }; class topper };

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[Page 3]

13. Which of the following combines two concurrent activities and re-introduces them to a flow where only time? (A) Fork symbol (C) Decision symbol

inheritance. 14. Class X, class Y and class Z are derived from class base. This is (A)

Multiple (C) Multilevel 15. Predict the output: class base{ int arr [10];}: class b1 :

public base }; class b2 : public base ; class derived : public b1, public b2 ;

(C) 0 main () {cout << size of (derived); return;} (A) 40

(A) template function Name (Parameters) (C) Both A and B

16. Which of the following best defines the syntax for template function?

17. Predict the output template <int i> void fun() {i-20; cout <<i;} int main () {

(A) 10 (C) compile error

(A) int (C) char *

fun<10>0; return 0; }

19. Predict the output

(B) Joint symbol (D) Note symbol

Page 3 of 6

1 activity can be performed at a

(B) Hierarchical (D) Single

18. Return type of uncaught exception () is

main () try (throw "b");}

(B) 80 (D) 4

return type (B) template

(A) Box

return type function name (parameters) (D) template return type

(C) UML packages

(B) 20 (D) 5


```
catch (int param) {cout<< "int exception";
```

(B) bool (D) double

```
catch (...) {cout << "default exception"; cout <<"After Exception";, return o;} (A) int  
exception after exception (B) (C) default exception (D) default
```

20. What is a collection of model elements called?

int exception exception

(B) Dependency

exception

(D) Package members

after

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[Page 4]

21. a. Construct a C++ program to create a class employee with data members as Employee_name, EID, phone, DoJ, no of hours-worked, pay rate. Write necessary member functions to calculate and display the salary of employee. (Use default value 10,000 for pay-rate). Apply the concept of default, parameterized and copy constructor to develop this complete code in C++,

PART -B (5 x 8 = 40 Marks) Answer ALL Questions

b. Rahul and Guldeep plays a mathematical game between each others. The game is all about

finding root of quadratic equations. The concept of constructor need to be used for getting input values. The constraints are $1 \leq a \leq 50$

Page 4 of 6

$1 \leq b \leq 50$ | $1 \leq c \leq 50$ Write a C++ program to get the described output for above scenarios.

22. a.i. Create a class time which contains

(OR)

Hours, Minutes, seconds. Write a C++ program using operator overloading for the following:

1) ==: To check whether two time are same or not. 2) >>: To accept the time 3) <<: To display the time ii. Draw sequence diagram for car rental system. Where the admin able to login into the system using their credentials? After login user can manage all the operations

passengers, drivers, cars, booking, car routes, payment gateway. Identify various objects involved in this project. Draw sequence and collaboration diagram for the same project.

Explain the components in detail.

or not. b.i. Write a program in C++ to check the given number is Armstrong number (OR)

A student is given a task to overload == operator. Armstrong number is a number that is equal to sum of cube of its digits. Input is an integer ranges $1 \leq N \leq 8000$. If the input is

Armstrong, print Armstrong number otherwise print not an Armstrong number. ii. Draw

collaboration diagram for the following scenario. *students of a university have membership

in library. *Librarian verifies the membership account. *If the account is valid, then

student requests a book. *The librarian checks the availability of the book. *If the book is available, it will be issued to the student. *The Librarian processes the fine details.

Explain the components used.

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[Page 5]

23. a.1. Write a program to calculate the total marks of each students of a class in

physics, chemistry and maths and average marks of the class. The number of students in the class are entered by the user. Create a class named marks with data members for roll number,

name and marks. Create three other classes inheriting the marks class namely phy, che and

maths which are used to define marks in individual subject of each student. Implement the above scenario using inheritance concepts. 11. You are required to present online design of a system that will be used by doctors at a medical practice to keep patient records. The system should record each consultation between a doctor and a patient, any illness diagnosed by the doctor, and any drugs prescribed to the patient. At regular intervals, an auditor will use the system to check whether the same drug is bring prescribed repeatedly to a particular patient. Sketch a diagram. Explain the components method. (OR) b.i. Construct a C++ program using class and object to implement Fibonacci series of given number using recursion concept and abstract class concept but the condition is, program has to refer function in cache memory and not from main memory. ii. Construct statechart diagram for accessing payroll details for particular employee. 24. a.i. Write a program to demonstrate the use of special functions, constructor and destructor in class template. The program is used to find the biggest of two entered numbers. ii. Supply code for computeAge () function such that an exception of type invalid input exception is thrown whenever either of two inputs is negative or if the current year is less than date of birth. (OR) b.i. Write a program to define function template for calculating square of given numbers with different datatypes. ii. Implement a main function such that, enclosed within a try statement, the computeAge) function is executed with invalid inputs. A catch statement should catch the exception, print out an error message and cause the program to immediately exit. 25. a. Write a C++ code to demonstrate sorting of 2D vector on basis of number of columns in ascending order.

Page 5 of 6

(OR) b. Write a C++ code using fstream class object to write the content in file and then to read the content of file.

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PART -C(1x 15 = 15 Marks) Answer ANY ONE Questions 26. Consider an electricity bill statement (EBS) real time application. The EBS calculates electricity charger (EC) from the units consumed by consumer. Construct class and usecase diagram for above diagram. 27. Create usecase and class diagram for library management system.

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Marks

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY Ramapuram Campus, Bharathi Salai, Ramapuram, Chennai
- 600089

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

QUESTION BANK

DEGREE / BRANCH: B.Tech-IT

III SEMESTER

SUB CODE - SUBJECT NAME

18CSC202J - OBJECT ORIENTED DESIGN AND PROGRAMMING

Regulation 2018

Academic Year 2022-23 (ODD SEMESTER)

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

Ramapuram Campus, Bharathi Salai, Ramapuram, Chennai-600089

DEPARTMENT OF INFORMATION TECHNOLOGY

QUESTION BANK

SUBJECT : 18CSC202J - OBJECT ORIENTED DESIGN AND PROGRAMMING SEM/YEAR: III/II Course

Outcomes CO- 1: Identify the class and build domain model CO-2 : Construct programs using
method overloading and operator overloading CO-3:

Create programs using inline, friend and virtual functions, construct programs using standard templates

CO-4 : Construct programs using exceptional handling and collections CO-5 : Create UML
component diagram and deployment diagram CO-6 : Create programs using object oriented
approach and design methodologies

UNIT I

Comparison of Procedural and Object Oriented Programming- OOPS and its features -

I/O Operations, Data Types, Variables, static - Constants, Pointers, Type Conversions -

Features: Class and Objects - UML Diagrams Introduction- Feature : Class and Objects -

Examples of Class and Objects- UML Class Diagram and its components- Class Diagram relations and

Multiplicity - Feature Abstraction and Encapsulation - Application of Abstraction

and Encapsulation - Access specifiers - public, private - Access specifiers - protected,

friend, inline - UML Use case Diagram, use case, Scenario - Use case Diagram objects and

relations - Method, Constructor and Destructor - Method, Constructor and Destructor PART-A

(Multiple Choice Questions) Q.No

Questions

Course Outcome

Competence BT Level

1 Which of the following explains Polymorphism? A) int func(int, int); Float

func1(float, float); B) int func(int); Int func(int); C) int func(float); Int new_func(); D)

int func(); Int new_func();

C01 2

2 Find how many bytes are occupied by the following data types in a 32-bit system. A) Type int B) Type long double C) Type float D) Type long

C01 4

3 Which of the following is an important role of a function? A. give a name to a block of code B. reduce program size C. accept arguments and provide a return value D. help organize a program into conceptual units

C01 4

[Page 3]

4 The Unified Modeling Language is A. a program that builds physical models. B. a way to look at the organization of a program C. the combination of C++ and FORTRAN D. helpful in developing software systems.

C01 1

5 In C++, a function contained within a class is called A. a member function B. an operator C. a class function D. a method

C01 1

6 What happens if the base and derived class contains definition of a function with same prototype? A. Compiler reports an error on compilation B. Only base class function will get called irrespective of object C. Only derived class function will get called irrespective of object D. Base class object will call base class function and derived class object will call derived class function

C01 4

7 Which one of the following option is correct about the statement given below? The compiler checks the type of reference in the object and not the type of object A. inheritance B. Polymorphism C. Abstraction D. Encapsulation

C01 4

8 Which of the following functions are performed by a constructor? A) Construct a new class B) Construct a new object C) Construct a new function D) Initialize objects

C01 4

9 Which of the following is the correct class of the object cout? A) iostream B) istream C) ostream D) ifstream

C01 2

10 In UML, diagrams which capture system static structure and provide foundation for other models is called A) Deployment diagram B) Class diagram C) Component diagram D) Object diagram

C01 2

11 Find the error produced by compiler when private members are accessed? A) Can't access private message B) Code unreachable C) Core dumped D) Bad code

C01 4

[Page 4]

12 Choose the default access specifier for the class member A) public B) private C) protected D) None of the above

C01 1

13 Which of the following is CPP style type-casting?A) per=total/ (float)mB)

per=total/float(m)C) per = (float)total/mD) None of these

C01 1

14 What is the output of the following program?#include<iostream>using namespace std;void main(){char s() = "SRM";*s = 'R';cout<<s<<endl;}A) RRMB) SRMC) SRRD) None of these

C01 4

15 What does the following statement mean?int (*fp)(char*)

A) pointer to a pointer B) pointer to an array of chars C) pointer to function taking a char* argument and returns an int D) function taking a char* argument and returning a pointer to int

C01 4

16 Which of the following concepts of OOPS means exposing only necessary information to client? A) Encapsulation B) Abstraction C) Polymorphism D) Data binding

C01 2

17 Which of the following is illegal? A) int *ip; B) string s, *sp = 0;C) int i; double* dp = &i; D) int *pi = 0;

C01 4

18 Which member can never be accessed by inherited classes?A) Private member function B) Public member function C) Protected member function D) All can be accessed

C01 1

[Page 5]

19 Analyze the code and choose the correctint a=100, b=200;int *p=&a, *q=&b;p=q;

A) b is assigned to a B) p now points to b C) a is assigned to b D) q now points to a

C01 4

20 Mention the size_t integer type in C++ is? A) Unsigned integer of at least 64 bits B) Signed integer of at least 16 bits C) Unsigned integer of at least 16 bits D) Signed integer of at least 64 bits

C01 2

21 Which among the following is not a property of an object?a) Identityb) Propertiesc) Attributesd) Names

C01 1

22 Which is most appropriate comment on following class definition?class Student{ int a;public : float a;};a) Error : same variable name can't be used twiceb) Error : Public must come firstc) Error : data types are different for same variabled) It is correct

C01 5

23 Instance of which type of class can't be created?a) Anonymous classb) Nested classc) Parent classd) Abstract class

C01 2

[Page 6]

24 What is the output of following code?int n=10; // globalclass A(){ private : int n;public : int m;A(){ n=100; m=50;}void disp(){ cout<<"n"<<m<<n;};a) 1050100b) 1005010c) n5010d) n50100

C01 5

25 Find the output of the following program.

```

class education{ char name[10];public : disp(){
cout<<"Its education system";}
}
class school:public education{ public: void dsip(){ cout<<"Its
school education system";}};
void main(){ school s;s.disp();}}
a) Its school education
system
b) Its education system
c) Its school education system
d) Its
education system
e) Its school education system

```

C01 5

PART B (4 Marks)

1 How can we prevent a class from instantiation? C01 22 Construct Use-case diagram for an Online Shopping Application C01 33 List out the difference between procedure oriented programming & Object oriented programming

C01 1

4 Write syntax of class, objects and methods and explain with example.

C01 2

[Page 7]

5 Write a C++ program to generate factorial of a number using class C01 36 Write an example program to demonstrate type conversions and explain

C01 2

7 Consider a Banking System. Identify three entities in the system which can be represented using classes and show the relationship between them using UML class diagrams

C01 6

8 What is a constructor? What are its uses? C01 2

PART C (12 Marks)

1 Write the problem statement for Library Management system. Design UML Class diagram and explain its components

C01 6

2 There are 50 computers available in computer programming lab where each computer is used six hours per day. Write a C++ program using classes and objects that contains getDetail() for getting input from user, calculateSecondPerDay() for calculating the usage of each computer in seconds per day, calculateMinutesPerWeek() for calculating the usage of each computer in minutes per week, calculateHourPerMonth() for calculating usage of each computer in hour per month and calculateDayPerYear() for calculating usage of each computer in day per year

C01 6

3 Give example for cast? Explain OOPS features with suitable example.

C01 2

4 A University conducts examinations and the results are announced. Prepare a report for the following:

- Print the marks in the register number order semester wise for each department.
- Print the Arrear list semester wise.
- Prepare a Rank list for each department.

C01 3

5 Create three classes with names Shape, Rectangle and Circle and make use of the functions getData(), printData(), and area(). To find the area of circle and rectangle, which type of inheritance is suitable? Why? Explain?

C01 6

UNIT II Types of constructor (Default, Parameter), Static and copy constructor - Feature

Polymorphism: Constructor overloading, Method Overloading -

Example for method overloading, Method Overloading: Different parameter with different return values

- Operator Overloading and types, Overloading Assignment Operator - Overloading Unary Operators,

Example for UnaryOperator overloading- Overloading

BinaryOperators,ExampleforBinaryOperatoroverloading- UMLInteractionDiagrams, Sequence Diagram-CollaborationDiagram,ExampleDiagram-Feature:Inheritance, Inheritance and its types
PART-A (Multiple Choice Questions)Q.No

Questions

CourseOutcome

CompetenceBT Level

1 While overloading binary operators using member function, it requires___ arguments.a.

Zerob.Onec. Twod. Three

C02 BT1

[Page 8]

2 Which of the followings are true about constructors?A. A class can have more than one constructor.B. They can be inherited.C. Their address can be referred.D. Constructors cannot be declared in protected section of the class.E. Constructors cannot return values.a. Only A, B, Db. A,B,D,Ec. A,C,Ed.A,D,E

C02 BT1

3 Which of the following keyword is used to overload an operator?a.

overloadb.operatorc.friendd.override

C02 BT1

4 What will happen if a class is not having any name?a. it cannot have a destructorb. It cannot have a constructor.c. It is not allowed.d.Both A and B

C02 BT1

5 Which inheritance type is used in the class given below?class A : public X, public Ya. Multilevel inheritanceb. Multiple inheritancec. Hybrid inheritanced .Hierarchical Inheritance

C02 BT1

6 Which of the following operators cannot be overloaded?a. []b. ->c.?:d. *

C02 BT1

7 In which of the following a virtual call is resolved at the time of compilation?a. From inside the destructor.b. From inside the constructor.c. From inside the main ().d.Both A and B

C02 BT2

8 Which of the following operator is overloaded for object cout?a. >>b. <<c. +d. =

C02 BT1

9 Assume class TEST. Which of the following statements is/are responsible to invoke copy constructor?a. TEST T2 (T1)b. TEST T4 = T1c. T2=T1d.Both A and B

C02 BT2

[Page 9]

10 Which of the following is the perfect set of operators that can't be overloaded in CPP?a. +=, ?, ::, >>b. >>, <<, ?,*,sizeof()c. ::, ., .*,?:d. ::, ->, *, new delete

C02 BT2

11 How many operators are supported by C++?a. 30 operatorsb. 40 operatorsc. 45 operatorsd. 65 operator

C02 BT1

12 A non-member function that is given access to all members of a class within it is declared, is called a. Access function b. Friend function c. Operator function d. None of them
C02 BT1

13 Which of the following operators should be preferred to overload as a global function rather than a member method? a. Postfix ++ b. Comparison Operator c. Insertion Operator << d. Prefix ++
C02 BT1

14 We can overload which of the following C++ operators a. Arithmetic operator (+, -, *, /) b. Class Member Access Operators (., .*) c. Size operator (sizeof) d. Conditional operator (?:)
C02 BT2

15 Operator overloading is also called polymorphism a. run time b. initial time c. Compile time d. Completion time
C02 BT1

16 Operator overloading is done with the help of a special function called, which describes the special task of an operator. a. overloading function b. special task function c. detail function d. operator function
C02 BT1

17 Overload an operator by naming it a. variable b. built-in type c. function d. class
C02 BT1

18 Which of the function operator cannot be over loaded a. <= b. ?: c. == d. *
C02 BT2

[Page 10]

19 Kind of diagrams which are used to show interactions between series of messages are classified as a. activity diagrams b. state chart diagrams c. collaboration diagrams d. object lifeline diagrams
C02 BT1

20 Dynamic aspects related to a system are shown with help of a. sequence diagrams b. interaction diagrams c. deployment diagrams d. use case diagrams
C02 BT1

21 Determine which diagrams are used to show interactions between series of messages a. Activity diagrams b. State Chart diagrams c. Collaboration diagrams d. Object Lifeline diagrams
C02 BT1

22 Identify the syntax of overloading operator + for class A? a. A operator + (arg_list){} b. A operator [+] (arg_list){} c. int +(arg_list){} d. int [+] (arg_list){}
C02 BT1

23 Classify three different types of message arrows a. Synchronous, asynchronous, asynchronous with instance creation b. Self, multiplied, instance generator c. Synchronous, asynchronous, synchronous with instance creation d. None of the above
C02 BT1

24 Which feature of OOP indicates code reusability? a. Abstraction b. Polymorphism c. Encapsulation d. Inheritance
C02 BT1

25 For constructor overloading, each constructor must differ in _____ and

_____a. Number of arguments and type of argumentsb. Number of arguments and return typec. Return type and type of argumentsd. Return type and definition

C02 BT1

PART B (4 Marks)

1 What is the necessity of constructor overloading? C02 BT22 Categorize the types of Constructors C02 BT13 Define method overloading. Write a program to implement methodoverloading with different number of arguments and same return types

C02 BT3

4 Write down the restrictions on Operator overloading C02 BT25 Can we have virtual destructors? If so what is the use of virtualdestructors.

C02 BT3

6 Define collaboration diagram with its notation C02 BT1

[Page 11]

7 Explain the modes of inheritance with an example C02 BT28 Judge the output of the following C++ code?

```
#include <iostream>using namespace std;class Integer{ int i;public:Integer(int ii) : i(ii) {}const Integeroperator+(const Integer&rv) const{cout<< "operator+"<<endl;return Integer(i + rv.i);}Integer&operator+=(const Integer&rv){cout<< "operator+="<<endl;i += rv.i; return *this;} };int main(){int i = 1, j = 2, k = 3;k += i + j;Integer ii(1), jj(2), kk(3); kk += ii + jj;}
```

C02 BT4

9 Relate the differences between constructors and destructors C02 BT2

10 Define overloading unary and binary operators with an example C02 BT2

PART C (12 Marks)

1 Define UML Sequence diagram with its notations. Draw the sequencediagram for Online hotel management

C02 BT4

2 Write a C++ program for Constructor overloading (parameterized, defaultand copy)

C02 BT3

3 Define Sequence and Collaboration diagram. Draw the sequence andcollaboration diagram for ATM amount withdrawal

C02 BT4

4 Define interaction diagram. Draw an Sequence diagram for onlinehospital management

C02 BT4

5 Explain in detail about operator overloading and its types with example C02 BT3

UNIT IIIFeature Inheritance: Single and Multiple, Inheritance: Multilevel, Hybrid, Hierarchial-AdvancedFunctions: Inline,Friend- Advanced Functions: Virtual, Overriding- Advanced Function: Pure Virtual function- Example for Virtualand pure virtual function- Abstract class and Interface- UML State Chart Diagram- UML Activity DiagramPART-A (Multiple Choice Questions)Q.No

Questions

CourseOutcome

CompetenceBT Level

[Page 12]

1 Diagrams which are used to distribute files, libraries and tables across topology of hardware are called?
a) deployment diagrams
b) use case diagrams
c) sequence diagrams
d) collaboration diagrams
Ans: d

C04 BT 1

2 Name the function whose definition can be substituted at a place where its function call is made:
a) friends function
b) inline function
c) volatile function
d) external function
Ans: b

C04 BT 2

3 Activity diagram, use case diagram, collaboration diagram and sequence diagram are considered as types of
a) non-behavioral diagrams
b) nonstructural diagrams
c) structural diagrams
d) behavioral diagrams
Ans: d

C04 BT 2

4 Use of pointers or reference to an abstract class gives rise to which among the following feature?
a) Static Polymorphism
b) Runtime polymorphism
c) Compile time Polymorphism
d) Polymorphism within methods
Ans: b

C04 BT 3

5 Which diagram in UML shows a complete or partial view of the structure of a modeled system at a specific time?
a) Sequence Diagram
b) Collaboration Diagram
c) Class Diagram
d) Object Diagram
Ans: d

C04 BT 2

6 Can abstract class have main () function defined inside it?
a) Yes, depending on return type of main()
b) Yes, always
c) No, main must not be defined inside abstract class
d) No, because main() is not abstract function
Ans: b

C04 BT 1

7 If there is an abstract method in a class then, _____
a) Class must be abstract class
b) Class may or may not be abstract class
c) Class is generic
d) Class must be public
Ans: a

C04 BT 1

[Page 13]

8 Which of the following UML diagrams has a static view?
a) Collaboration
b) Use case
c) State chart
d) Activity
Ans: b

C04 BT 1

9 11. Which keyword is used to declare the friend function?
a) friend
b) friendc)
classfriend
d) myfriend
Ans: b

C04 BT 2

10 Which of the following cannot be used to declare a class as a virtual?
a) Methods
b) Properties
c) Events
d) Fields
Ans: d

C04 BT 2

11 Which of the given modifiers can be used to prevent Method overriding?
a) Static
b) Constant
c) Sealed
d) final
Ans: c

C04 BT 1

12 Which problem arises due to multiple inheritances, if hierarchical inheritance is used previously for its base classes?
a) Diamond
b) Circle
c) Triangled
d) Loop
Ans: a

C04 BT 2

13 How many classes should a program contain to implement the multiple inheritance?
a) Only

1b) At least 1c) At least 3d) Exactly 3Ans: c

C04 BT 1

14 How many basic types of inheritance are provided as OOPfeature?a) 4b) 3c) 2d) 1Ans: a

C04 BT 1

[Page 14]

15 How can you make the private members inheritable?a) By making their visibility mode as public onlyb) By making their visibility mode as protected onlyc) By making their visibility mode as private in derived classd) It can be done both by making the visibility mode public orprotectedAns: d

C04 BT 1

16 Which access type data gets derived as private member in derived class?a) Privateb) Publicc) Protected d) Protected and PrivateAns: a

C04 BT 1

17 Which programming language doesn't support multiple inheritances?a) C++ and Javab) C and C++c) Java and Small Talkd) JavaAns: d

C04 BT 2

18 Which among the following best defines single level inheritance?a) A class inheriting a derived classb) A class inheriting a base classc) A class inheriting a nested classd) A class which gets inherited by 2 classesAns: b

C04 BT 2

19 Is it compulsory to have constructor for all the classes involved in multiple inheritance? a) Yes, alwaysb) Yes, only if no abstract class is involvedc) No, only classes being used should have a constructord) No, they must not contain constructorsAns: b

C04 BT 2

20 Can the derived class be made abstract if multiple inheritance is used?a) No, because other classes must be abstract too b) Yes, if all the functions are implementedc) Yes, if all the methods are predefined d) No, since constructors won't be thereAns: d

C04 BT 2

21 Which keyword is used to declare the friend function?a) friendb) friendc) classfriendd) myfriendAns: d

C04 BT 1

[Page 15]

22 Pick out the correct option.a) We cannot make an instance of an abstract base classb) We can make an instance of an abstract base classc) We can make an instance of an abstract super classd) We can make an instance of an abstract derived classAns: a

C04 BT 1

23 Which is used to create a pure virtual function?a) \$b) =0c) &d) !Ans: b

C04 BT 3

24 Which is also called as abstract class?a) virtual functionb) pure virtual functionc) derived classd) base classAns: b

C04 BT 3

25 What is the syntax of friend function?a) friend class1 Class2;b) friend class;c) friend classd) friend class()Ans: a

C04 BT3

PART B (4 Marks)

1 Explain Virtual Function

C02 BT2

2 What is pure virtual function?

C02 BT2

3 Explain the notations of Activity Diagram and Statechart Diagram

C04 BT3

4 What is Inheritance? Benefits of Inheritance?

C02 BT2

5 Describe the types of inheritance?

C02 BT3

6 Difference between Virtual Function and Pure Virtual Function.

C02 BT2

7 Explain about Abstract Class and Interface.

C03 BT2

8 What is friend function and friend class?

C03 BT2

PART C (12 Marks)

1 Draw UML state chart and Activity Diagram for ATM Machine

C04 BT4

2 Describe in detail about advanced friend function and friend class with example?

C02 BT2

3 Describe Multilevel Inheritance with example program?

C02 BT3

4 Explain Pure Virtual Function with example program?

C03 BT2

[Page 16]

5 Draw UML state chart and Activity Diagram for Library Management System

C04 BT4

UNIT IV Generic - Templates : Introduction- Function Template

- Example programs: Function Templates- Class Template- Example programs : Class Templates

- Exceptional Handling: try and catch- Multilevel exceptional- throw and throws - finally-

User defined exception- Example programs using C++- Dynamic modelling: Package Diagram- UML Component diagram - Deployment Diagram - Example : Package, Deployment, Component diagram. PART-A

(Multiple Choice Questions) Q.No

Questions

Course Outcome

Competence BT Level

1 The STL can be used as a standard approach for-----a) Storing and sorting b) Storing and processing data c) data processing only d) storing only

C03 1

2 Name the Container which uses both stack and queue. a) storage b) linked list c) queueing d) Deque

C03 1

3 Identify the characteristics of vector container.a) Relocating, expandable arrayb) Fixed sized) Doubly linked listd) link vector

C03 2

4 Associative container uses-----to access data.a) queueb) Keysc) stackd) string

C03 1

5 Class templates are generally used for-----a) Data storageb) debugc) fixed data typed) storage

C03 1

6 In UML, Templates are also called as-----a) containerb) Modifiedc) Parameterizedd) generic

C03 1

7 Identify the validity of template parameters?a) inside that block onlyb) inside the classc) whole programd) inside the main class

C03 2

[Page 17]

8 Identify which among the following is not correct.a)template <class T> func(T x) {}b) template <class T> class myObject {}c) template <class T> class myObj { template <class R> memFunc() {};}d)All of the above are correct

C03 2

9 Examine whether templates are conceptually related to polymorphism?a) Not Relatedb) Only when the template types are objectsc) Yes, but compile-time polymorphismd) Yes, but run-time polymorphism

C03 2

10 Identify an invalid template declaration.[L2, R5-683]a) template <int x> int func() {return x;}b) template <double x> double func() {return x;}c) template <typename x> void func(x t) {}d) It is not possible in CPP to restrict a function

C03 2

11 Explore the correct statement about string template?a) It is used to replace a string.b) It is used to replace a string with another string at runtime.c) It is used to delete a string.d) None of the above

C03 2

12 _____type of program can be included in tryblock?a) static memory allocationb)dynamic memory allocationc) const referenced) pointer

C04 1

13 -----statement is used to catch all types of exceptions.a) catch()b) catch(Test t)c)catch(...)d) no one of the mentioned

C04 1

14 The ----- class name must be included in the class in which it is located.a) tryb)Exceptionc) catchd) template

C04 1

15 Select the ways to represent nodes in a deployment diagram?a) Nodes instances are underlined identifiers of the form name:typeb) The name may be left off, indicating an unnamed instance of the typec) The type may be left off, indicating a named instance with

an unspecified type) All of the mentioned

C05 1

16 ____ specifies additional detail about UML element. a) Stereotype b) container c) associative container d) data processing

C05 1

[Page 18]

17 ----- is visible only to its containing package and to its nested a) package b) protected c) public d) Private package

C05 1

18 Notation is used to specify the required and provided interfaces of the components. The interfaces between the components are named as ---- a) Assembly connectors b) cooling controllers c) Environmental controller d) Plan analyst

C05 1

19 List the 3 essential elements of a deployment diagram. a) Artifacts, nodes and connections. b) stack, queue, deque c) memory, database, connections d) package, element, deployment

C05 1

20 Activity, use case diagram, collaboration diagram and sequence diagram are categorized as _____. a) non-behavioral diagrams b) non structural diagrams c) structural diagrams d) Behavioral diagrams

C05 1

21 Recognize which diagram is used to distribute files, libraries and tables across topology of hardware? a) Deployment diagrams b) use case diagrams c) sequence diagrams d) collaboration diagrams

C05 2

22 List the essentials in package diagram a) Package notation, element visibility, dependency relationship b) package notation, sequence, dependency relationship c) Dependency, element visibility d) package, deployment, sequence

C05 1

23 Good packages are ----- coupled and highly cohesive among the elements in package. a) Tightly b) highly c) loosely d) semi

C05 1

[Page 19]

24 Identify the core element of UML in the below figure?

a) Node b) Interface c) Class d) Component

C05 2

25 Recognize the UML diagram shown below?

a) Component b) Deployment c) Use case d) DFD

C05 2

PART B (4 Marks) 1 What do you mean by Generic Programming? What are its advantages and state few applications?

C03 1

2 Define a Class Template. Write a suitable example program. C03 23 What is a Function

Template? Illustrate with a suitable example program.

C03 2

4 What do you mean by Overloaded Function Template? What are the rules to be followed to select a suitable template?

C03 1

5 Distinguish between overloaded functions and function templates C03 2

6 What is the need for template function in C++? What are the advantages?

C03 1

7 Give the differences between Class template and Function template. C03 28 What is an exception? How it is handled in C++? C04 1

9 Write a program to demonstrate the concept of rethrowing an exception.

C04 3

10 What are the two kinds of exception? C04 1

11 Illustrate multiple catch statements with a suitable example. C04 2

12 When should a function throw an exception? Give an example to illustrate it.

C04 2

13 What is `uncaught_exception()` function? why do we need it? C04 214 When do we need multiple catch Handlers? Give an example. C04 2

[Page 20]

15 What are standard exceptions? List the types of exception and specify the position when it was generated?

C04 1

16 What is a component diagram and state its artifacts. C05 1

17 Define Deployment diagram. State the artifacts to be identified before drawing a Deployment diagram.

C05 1

18 What are the uses of the Component diagram and Deployment diagram?

C05 2

19 Give the notations of Component diagram and Deployment Diagram C05 1

20 Define a Package Diagram. Give the advantages of using a Package diagram.

C05 1

PART C (12 Marks)1 Discuss in detail on Class Template with a suitable example. C03 22 What is a Function Template? Discuss in detail with a suitable program. C03 23 Write a program which generate a template class by which one can perform integer type data addition and float type data addition.

C03 3

4 Discuss in detail on Overloading Function template. Illustrate it with a suitable program.

C03 2

5 Explain how the Class Template can be used with Operator overloading with a program.

C03 2

6 Write a C++ program to develop a Simple Calculator to perform arithmetic operations using Class Template.

C03 3

7 What is an exception? How it is handled in C++ programs? Explain how the control is transferred when exceptions occur during program execution. Write a program to illustrate exception handling.

C04 2

8 Write a program to show how to restrict the types of exceptions that can be thrown by a function.

C04 3

9 Write a program to show how to rethrow an exception. C04 310 Write a C++ program to demonstrate the use of try, catch, throw and nested try.

C04 3

11 What is a user defined exception. Write down the scenario where we require user defined exceptions.

C04 2

12 When do we need multiple catch blocks for a single try block? Write a program to illustrate it.

C04 2

[Page 21]

13 Explain in detail on the use of Multiple catch statements in a program with a suitable example. Discuss the importance of "catch all exception" with a program.

C04 2

14 Explain Deployment Diagram with a suitable example. C05 215 What is a Package Diagram. Illustrate it with a suitable example. C05 216 Draw the Component, Deployment and package diagram for the ATM Banking system. Explain the system with the notations used in each diagram.

C05 3

17 Design the Component, Deployment and package diagram for the Airline Reservation system. Explain the system with the notations used in each diagram.

C05 3

18 Explain the Component, Deployment and package diagram for the Course Registration system with a neat diagram. Illustrate the system with the notations used in each diagram.

C05 3

19 Discuss in detail on Component Diagram with a suitable example. C05 220 Draw the Component, Deployment and package diagram for the Online Shopping system. Explain the system with the notations used in each diagram.

C05 3

UNIT V STL: Containers, sequence and Associative containers, Sequence Containers: Vector, List, Deque, Array, STL - stack - Associative Containers - Map - MultiMap - Iterator and Specialized Iterator - Functions of Iterator - Algorithms - find() - count() - sort() - search() - merge() - function object: for_each, transform - streams and Files : introduction - classes and errors - disk file handling - reading and writing PART-A (Multiple Choice Questions) Q.No Questions

Course Outcome

Competence BT Level

1 What kind of library is Standard Template Library? a) Polymorphic b) Generic c) Both Polymorphic & Generic d) None of the mentioned

Ans:b

C06 BT1

[Page 22]

2 To what type of object does the container can be instantiated? a. intb. floatc. doubled.
any type of object

Ans:d

C06 BT1

3 What type of class template is list? a. Class-basedb. Node-basedc. Method-basedd. None of
the mentionedAns:b

C06 BT2

4 What type of access does deque and vector provide? a. Linear accessb. Parallel accessc.
Random accessd. None of the mentionedAns:c

C06 BT2

5 Where does the vector add the item? a. Endb. Insertc. Middled. None of the mentionedAns:a

C06 BT1

6 Which are not full container classes in C++?a. Sequence containerb. Associative
containerc. Container adaptord. None of the mentionedAns:c

C06 BT1

7 What is the lifetime of the element in container?a. Whole programb. Outside the blockc.
Everywhered. Only on that containerAns:d

C06 BT1

8 Which operator is used to insert the data into file? a. >>b. <<c. <d. None of the
AboveAns: b

C06 BT1

[Page 23]

9 Which function is used to position back from the end of file object?a. seekgb. seekpc. both
seekg&seekpd. None of the AboveAns:a

C06 BT1

10 How many objects are used for input and output to a string?a. 1b. 2c. 3d. 4Ans:c

C06 BT2

11 Which is used to handle the exceptions in c++?a. catch handlerb. handlerc. exception
handlerd. None of the MentionedAns:c

C06 BT1

12 Which type of program is recommended to include in try block?a. static memory allocationb.
dynamic memory allocationc. const referenced. pointer

Ans:b

C06 BT1

13 Which statement is used to catch all types of exceptions?a. catch()b. catch(Test t)c.
catch(...)d. None of the AboveAns:c

C06 BT1

[Page 24]

14 What do Associate containers implement? a. Arraysb. Associative arraysc. Functional

Arraysd. Static arrays Ans: b

C06 BT1

15 By using which of the following the elements in the associate container can be efficiently accessed? a. Keyb. Positionc. Both Key & Positiond. Value Ans: a

C06 BT3

16 How many items are presented in the associate container? a. 2b. 3c. 4d. 5 Ans: c

C06 BT1

17 What are the containers? a. Containers store objects and datab. Containers stores all the algorithmsc. Containers contain overloaded functionsd. Containers contain set of IteratorsAns: a

C06 BT1

18 In how many categories, containers are divided? a. 1b. 2c. 3d. 4

C06 BT2

[Page 25]

19 What are the Sequence Containers? a. Containers that implements data structures which can be accessed sequentiallyb. Containers that implements sorted data structures for fast search in $O(\log n)$ c. Containers that implements unsorted(hashd) data structures for quick search in $O(1)$ d. Containers that implements data structures which can be accessed non-sequentiallyAns: a

C06 BT2

20 How many Sequence Containers are provided by C++? a. 2b. 3c. 4d. 5Ans: d

C06 BT1

21 What is the Standard Template Library? a) Set of C++ template classes to provide common programming data structures and functionsb) Set of C++ classesc) Set of Template functions used for easy data structures implementationd) Set of Template data structures onlyAns: a

C06 BT1

22 What are Unordered Associative Containers? a) Containers that implements data structures which can be accessed sequentiallyb) Containers that implements sorted data structures for fast search in $O(\log n)$ c) Containers that implements unsorted(hashd) data structures for quick search in $O(1)$ d) Containers that implements data structures which can be accessed non-sequentiallyAns: c

C06 BT1

23 What are Iterators? a) Iterators are used to iterate over C-like arraysb) Iterators are used to iterate over pointersc) Iterators are used to point memory addresses of STL containersd) Iterators are used to iterate over functionsAns: C

C06 BT1

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24 Which header file is used for Iterators? a) <iterator>b) <algorithm>c) <iter>d) <loopIter>Ans: a

C06 BT1

25 Consider that the variable str is of type std::string. What is the correct way to get the C-style string from str? A. Cast str to const char* as in ((const char*)&str) B. Use

str.get_c_style_string()C. Use str.c_str()D. Use str.data()Ans:C

C06 BT2

PART B (4 Marks)1 What are containers in C++ STL? C06 BT12 What are the 3 entities of STL in C++? C06 BT13 What is true about his statement in C++?

```
std::vector<int> vecInts(5);
```

C06 BT2

4 Justify your answers Is it possible to initialize any Vectorwith an Array in C++?

C06 BT1

5 Difference betweenVector Vs List C06 BT26 Different Ways to Initialize a List and elaborate them C06 BT27 How toerase elements from List using Iterators C06 BT18 How toRemove Elements from a List while Iterating C06 BT1PART C (12 Marks)1 Givesyntaxof andexplainvarious functions relatedto ifstream and ofstream classes:

```
seekp(),getline(),hide(),tail().
```

C06 BT3

2 Explain the use of ifstreamandofstreamclasses forfile input and output.

C06 BT33 Explain the file operation functions in C++ tomanipulate the position of file pointers inarandomaccess file.

C06 BT3

4 What is the purpose of push_back(), push_front(),pop_back() and pop_front() functions of a list.

C06 BT2

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5 What does this function do?

```
void func() {  
std::vector<std::string> vecOfString(5, "Hi");for (std::string str : vecOfString)std::cout <<  
str << std::endl;}
```

C06 BT2

Note:

1. BT Level -Blooms Taxonomy Level

2. CO - Course Outcomes

BT1 -Remember BT2 - Understand BT3 - Apply BT4 - Analyze BT5 - Evaluate BT6 - Create